

A PROOF OF GLUCK'S CONJECTURE

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ABSTRACT. For a finite group H , let $\nu(H)$ denote the maximum order of a nilpotent subgroup of H . We prove that every finite solvable transitive permutation group P on a finite set Ω has a subset $\Delta \subseteq \Omega$ such that $|P : P_\Delta| \geq \nu(P_\Delta)$. We also prove that if a finite solvable group H acts faithfully and completely reducibly on a finite module V , then some $x \in V$ satisfies $|H : H_x| \geq \nu(H_x)$. Consequently, we settle Gluck's well-known conjecture, open since 1985: every finite solvable group G satisfies $|G : \mathbf{F}(G)| \leq b(G)^2$, where $\mathbf{F}(G)$ is the largest normal nilpotent subgroup of G and $b(G)$ is the largest degree of an irreducible complex character of G .

1. INTRODUCTION

In 1985, Gluck proposed the following conjecture [10].

Conjecture 1.1 (Gluck). Let G be a finite solvable group. Then $|G : \mathbf{F}(G)| \leq b(G)^2$, where $b(G)$ is the largest degree of an irreducible complex character of G .

The index $|G : \mathbf{F}(G)|$ measures how far G is from being nilpotent, whereas $b(G)$ is a representation-theoretic invariant. Conjecture 1.1 asserts that the two are closely tied. Our main result confirms the conjecture.

Theorem 1.2. *Conjecture 1.1 is true.*

Conjecture 1.1 was previously established in several cases: for groups of odd order by Espuelas [6], for solvable groups with abelian Sylow 2-subgroups by Dolfi and Jabara [5], for solvable $3'$ -groups by Yang [28], for groups in which $|\mathbf{F}(G)|$ is coprime to 6 by Cossey, Halasi, Maróti and Nguyen [2], for solvable S_4 -free groups by Meng, Ballester-Bolínches and Esteban-Romero [17], and for a class of iterated wreath products by Meng and Guo [19]. Here S_4 -free means having no section $H/K \cong S_4$, where $K \trianglelefteq H \leq G$. Related questions involving average character degrees were studied by Moretó [20].

The right-hand side of Conjecture 1.1, the largest degree $b(G)$ of an irreducible complex character, is itself a subtle invariant. Even for symmetric groups, the largest irreducible character degrees are not fully understood and have recently been investigated computationally by Craven [3]. For solvable groups, Section 2 shows that Clifford theory connects bounds of the form $|G : \mathbf{F}(G)| \leq b(G)^c$ with estimates for their orbits on faithful completely reducible modules. Large orbits, and especially regular orbits, have therefore played an important role. See, for example, [4, 15, 18, 26]. The problem of regular orbits for solvable primitive linear groups was studied by Yang and his collaborators [27, 29, 31]. We call a faithful irreducible quasi-primitive linear action of a solvable nonmetacyclic group *residual* if it has no regular orbit. The classification of residual actions used here is proved by Holt and Yang [12]. The companion problem of regular orbits on the *power set* of a permutation domain, that is, of subsets with trivial setwise stabiliser, was studied by Gluck [9, 11], by Seress [24] and by Zhang [33], among others. Such results enter the imprimitive linear case through the induced permutation action on the summands.

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Before the present work, the exponent in the best known general bound had been reduced in stages: to $13/2$ by Gluck in [10], to 3 by Moret  and Wolf [21] by way of Seress’s theorem that a primitive solvable permutation group has a base of size at most four [23], and to $\log(6 \cdot 24^{1/3})/\log 3 \approx 2.595$ by Yang [30].

Gluck identified the obstruction to proving the bound in Conjecture 1.1 in the very paper that posed the conjecture, where he wrote that his methods were “not strong enough to give best possible bounds, even for solvable groups” [10, p. 443]. More precisely, if a solvable group H acts faithfully and completely reducibly on a finite module V , the approach would require a vector $v \in V$ such that $|H_v| \leq |H|^{1/2}$, equivalently $|v^H| \geq |H|^{1/2}$. But the imprimitive action of $\mathrm{GL}(2, 2) \wr S_4$ on $\mathbb{F}_2^8$ “shows that this stronger version of Theorem A does not hold” [10, p. 447], since its largest orbit has length less than the square root of the group order. Meng and Guo [19, Lemma 8(2)] construct a subgroup of $\mathrm{GL}(4, 3)$ of order 1 152 whose orbits on $\mathbb{F}_3^4$ have lengths 1, 32, 24 and 24. Thus no argument requiring an orbit of length at least the square root of the group order can prove the conjecture in general. Yang also observed in [30, Remark 4.6] that examples of this kind suggest that an orbit theorem alone may not suffice to prove Gluck’s conjecture. We therefore abandon the search for a single large orbit. The reduction of Section 2 shows that Conjecture 1.1 follows from the weaker assertion that V always contains an *abundant* element, as defined below.

Notation and terminology. We use standard notation from group theory and character theory, as in [14]. All groups and modules are finite, and modules are written additively. For a group H we write $\pi(H)$ for the set of primes dividing $|H|$, $O_p(H)$ for the largest normal p -subgroup of H , $\mathrm{Irr}(H)$ for the set of irreducible complex characters, $\mathbf{F}(H)$ for the Fitting subgroup, $\Phi(H)$ for the Frattini subgroup, and $\mathbf{C}_H(\cdot)$ and $\mathbf{N}_H(\cdot)$ for centralisers and normalisers, respectively. We write $\nu(H)$ for the maximum order of a nilpotent subgroup of H . This invariant is denoted $|H|_{\mathrm{nil}}$ by Hung and Yang in their study of bounds involving nilpotent subgroups and largest character degrees [13]. When H acts on a set, H_Δ denotes the setwise stabiliser of a subset Δ , and Δ^c its complement. For a point x , $H_x := H_{\{x\}}$ is its stabiliser. Actions are written on the right, so $x^{gh} = (x^g)^h$. We set

$$\rho_H(\Delta) = \frac{|H : H_\Delta|}{\nu(H_\Delta)}$$

and $\rho_H(x) := \rho_H(\{x\})$. The quantities $\rho_H(\Delta)$ and $\rho_H(x)$ are the *abundance ratios* of Δ and x , respectively. A subset or point is *abundant* if its abundance ratio is at least 1. For any H -set, an H -orbit is *regular* if the stabiliser of one (equivalently, every) point in the orbit is trivial. Accordingly, a vector v is regular if $H_v = 1$, and a subset Δ is regular if its setwise stabiliser H_Δ is trivial. Throughout, q denotes a prime power and $\mathbb{F}_q$ a field of order q . A transitive permutation group is *primitive* if its only invariant partitions are the partition into singletons and the partition consisting of the whole set. An irreducible linear group is *primitive* if it admits no nontrivial system of imprimitivity. An irreducible linear group $H \leq \mathrm{GL}(V)$ is *quasi-primitive* if every normal subgroup $N \trianglelefteq H$ acts homogeneously on V , that is, if V is a direct sum of mutually isomorphic irreducible N -modules. This is the linear meaning of quasi-primitivity throughout the paper. In this paper, Γ denotes a set of coordinate indices and Δ a subset of a permutation domain. For blocks $D_1, \dots, D_m$ or summands $W_1, \dots, W_m$, we write $I = \{1, \dots, m\}$ for the index set and regard the induced permutation group as acting on I , with D_i or W_i represented by i . Table 1.1 lists the notation and terminology used repeatedly below.

TABLE 1.1. Where recurring notation and terminology are introduced.

term	location
<i>residual action</i>	Introduction
<i>abundant</i> subset or point ($\rho \geq 1$)	Introduction; Section 2
<i>regular pair</i> — an ordered pair of points	Section 2
<i>coordinate</i> ; <i>coordinate set</i> ; <i>coordinate tuple</i> ; <i>colour</i>	Section 3
<i>local group</i> ; <i>top group</i> P ; <i>top factor</i>	Section 3
<i>coordinate chain</i> $L \trianglelefteq K \trianglelefteq J$; <i>bottom group</i> L	Section 3
<i>chain-abundant orbit</i>	Definitions 4.1 and 7.1
<i>cubic condition</i>	Section 5
<i>cubic pair</i>	Definition 5.3
<i>balanced pair</i> — an ordered pair of orbits	Definition 7.1

Definition 1.3. Let H be a group and V an H -module. We say that H acts *completely reducibly* on V , and that V is a *completely reducible* H -module, if every H -invariant subgroup of V has an H -invariant direct complement. Equivalently, each $O_p(V)$ is elementary abelian and completely reducible as an $\mathbb{F}_p H$ -module, and

$$V = \bigoplus_{p \in \pi(V)} O_p(V).$$

Thus every irreducible summand of V is a finite-dimensional $\mathbb{F}_p H$ -module for some prime p .

The heart of the paper is the following statement, which makes no mention of character degrees. That it implies Conjecture 1.1 is shown in Section 2.

Theorem 1.4. *Let V be a faithful completely reducible H -module for a solvable group H . Then V contains an abundant element.*

The proof of Theorem 1.4 uses Theorem 1.5, which gives an abundant subset in every solvable transitive permutation group. The proof of Theorem 1.5 requires finite calculations only for primitive solvable permutation groups of degree at most nine. In particular, that proof is independent of the classification of residual actions in [12] and of the finite calculations for primitive linear groups used later in the proof of Theorem 1.4. We write P , rather than H , for the permutation group in the following theorem, since linear and permutation actions often occur together later in the paper.

Theorem 1.5. *Every finite solvable transitive permutation group P on a finite set Ω has an abundant subset.*

Organisation of the paper. Section 2 shows that Theorem 1.4 implies Conjecture 1.1 and reduces the proof of Theorem 1.4 to faithful irreducible modules over prime fields. Section 3 establishes the coordinate chain inequality, the main quantitative tool for imprimitive modules. Sections 4 and 5 prove Theorem 1.5 and its weighted strengthening for solvable transitive permutation groups. Section 6 proves the abundance theorem for primitive modules, using the residual classification and finite verification in the nonmetacyclic case and a semilinear argument in the metacyclic case. Section 7 establishes the stronger local alternatives used in Section 8 to prove the abundance theorem for imprimitive modules. Section 9 completes the proof of Theorem 1.4 and hence of Conjecture 1.1. The appendix describes the finite calculations used in the proof and how to repeat them.

2. REDUCTION TO IRREDUCIBLE MODULES

Recall that a point x of an H -set is *abundant* if $\rho_H(x) \geq 1$. For an H -set X , we call $(x, y) \in X \times X$ a *regular pair* if $H_x \cap H_y = 1$. Equivalently, (x, y) is a regular point for the diagonal action of H on $X \times X$.

Lemma 2.1. *Let a finite group H act on a finite set X , and let $(x, y) \in X \times X$ be a regular pair. Then $|H : H_x| |H : H_y| \geq \nu(H_x) \nu(H_y)$, and in particular one of x, y is abundant.*

Proof. Since $H_x \cap H_y = 1$ we have $|H_x| |H_y| = |H_x H_y| \leq |H|$. Hence

$$\nu(H_x) \nu(H_y) \leq |H_x| |H_y| \leq |H|$$

and

$$|H : H_x| |H : H_y| = \frac{|H|^2}{|H_x| |H_y|} \geq |H|.$$

It follows that

$$\rho_H(x) \rho_H(y) = \frac{|H : H_x| |H : H_y|}{\nu(H_x) \nu(H_y)} \geq 1,$$

and as both factors are positive, $\rho_H(x) \geq 1$ or $\rho_H(y) \geq 1$. $\square$

The following lemma shows that Theorem 1.4 implies Conjecture 1.1.

Lemma 2.2. *If every faithful completely reducible module for a finite solvable group contains an abundant vector, then Conjecture 1.1 is true.*

Proof. We choose a counterexample G to Conjecture 1.1 of minimal order.

We claim that $\Phi(G) = 1$. Assume for a contradiction that $\Phi(G) \neq 1$. Then $G/\Phi(G)$ satisfies Conjecture 1.1 by the minimal choice of G . Moreover, every irreducible character of $G/\Phi(G)$ inflates to an irreducible character of G , whence $b(G/\Phi(G)) \leq b(G)$. Since $\mathbf{F}(G/\Phi(G)) = \mathbf{F}(G)/\Phi(G)$,

$$|G : \mathbf{F}(G)| = |G/\Phi(G) : \mathbf{F}(G/\Phi(G))| \leq b(G/\Phi(G))^2 \leq b(G)^2,$$

contrary to G being a counterexample to Conjecture 1.1. Hence $\Phi(G) = 1$.

Since $\Phi(K) \leq \Phi(G)$ whenever $K \leq G$, we have $\Phi(O_p(\mathbf{F}(G))) = 1$ for every $p \in \pi(\mathbf{F}(G))$. Hence every $O_p(\mathbf{F}(G))$ is elementary abelian, and consequently $\mathbf{F}(G)$ is abelian.

Now choose $H \leq G$ minimal by inclusion such that $G = \mathbf{F}(G)H$, and set $N = H \cap \mathbf{F}(G)$. Since $\mathbf{F}(G) \leq G$, we have $N \leq H$. We claim that $N \leq \Phi(H)$. Otherwise some maximal subgroup $L < H$ does not contain N . Then $H = NL$, and hence $G = \mathbf{F}(G)H = \mathbf{F}(G)L$, contrary to the choice of H . Since $\mathbf{F}(G)$ is abelian, it centralises N . It follows from $G = \mathbf{F}(G)H$ and $N \leq H$ that $N \leq G$.

Suppose that $N \neq 1$. Since $\Phi(G) = 1$, there is a maximal subgroup $M < G$ such that $N \not\leq M$. As $N \leq G$, maximality gives $G = NM$. Since $N \leq H$, Dedekind's modular law gives $H = H \cap NM = N(H \cap M)$. Since $N \leq \Phi(H)$, every element of N is a non-generator of H . The equality $H = N(H \cap M)$ therefore forces $H = H \cap M \leq M$, contrary to $N \not\leq M$. Therefore $N = 1$, so $G = \mathbf{F}(G) \rtimes H$ and $H \cong G/\mathbf{F}(G)$.

As $\mathbf{F}(G)$ is abelian, all its irreducible characters are linear. Under pointwise multiplication, $V := \text{Irr}(\mathbf{F}(G))$ is its character group. Conjugation by H makes V the dual of the H -module $\mathbf{F}(G)$. By a theorem of Gaschütz [8, Satz 7], $\Phi(G) = 1$ implies that $\mathbf{F}(G)$ is completely reducible under conjugation by G . Since $\mathbf{F}(G)$ is abelian, it acts trivially on itself by conjugation. Thus this action induces an action of $G/\mathbf{F}(G)$ on $\mathbf{F}(G)$. It follows that $\mathbf{F}(G)$ is a completely reducible $G/\mathbf{F}(G)$ -module and hence a completely reducible H -module. This H -action is faithful because $\mathbf{C}_G(\mathbf{F}(G)) = \mathbf{F}(G)$ and $H \cap \mathbf{F}(G) = 1$. By [16, Proposition 12.1], duality preserves faithfulness and irreducibility. Applying this to $\mathbf{F}(G)$ and its irreducible summands shows that V is a faithful and completely reducible H -module. The hypothesis of the lemma therefore gives an abundant character $\lambda \in V$ with

$$(2.1) \quad |H : H_\lambda| \geq \nu(H_\lambda).$$

Every element of $\text{I}_G(\lambda) = \mathbf{F}(G) \rtimes H_\lambda$ has a unique expression fh , where $f \in \mathbf{F}(G)$ and $h \in H_\lambda$. Define

$$\hat{\lambda} : \text{I}_G(\lambda) \longrightarrow \mathbb{C}^\times, \quad \hat{\lambda}(fh) := \lambda(f).$$

The uniqueness of this expression makes $\widehat{\lambda}$ well defined. For $f_1, f_2 \in \mathbf{F}(G)$ and $h_1, h_2 \in H_\lambda$, we have $\lambda(h_1 f_2 h_1^{-1}) = \lambda(f_2)$. It follows from this equality and the linearity of λ that the value of $\widehat{\lambda}((f_1 h_1)(f_2 h_2))$ is

$$\widehat{\lambda}((f_1 h_1 f_2 h_1^{-1})(h_1 h_2)) = \lambda(f_1(h_1 f_2 h_1^{-1})) = \lambda(f_1)\lambda(f_2) = \widehat{\lambda}(f_1 h_1)\widehat{\lambda}(f_2 h_2).$$

Thus $\widehat{\lambda}$ is a linear character of $I_G(\lambda)$ extending λ . Let $\eta \in \text{Irr}(H_\lambda)$, and inflate η to $I_G(\lambda)$ with $\mathbf{F}(G)$ in its kernel. By Gallagher's theorem [14, Corollary 6.17], $\widehat{\lambda}\eta$ is irreducible. Its induction to G is irreducible by the Clifford correspondence [14, Theorem 6.11], and has degree

$$|G : I_G(\lambda)| (\widehat{\lambda}\eta)(1) = |G : I_G(\lambda)| \eta(1) = |H : H_\lambda| \eta(1).$$

Taking η of largest degree gives

$$(2.2) \quad b(G) \geq |H : H_\lambda| b(H_\lambda).$$

Since G is a nontrivial solvable group, $\mathbf{F}(G) \neq 1$, and hence $|H_\lambda| \leq |H| = |G : \mathbf{F}(G)| < |G|$. The group H_λ is solvable, so the minimality of G gives $b(H_\lambda)^2 \geq |H_\lambda : \mathbf{F}(H_\lambda)|$. Squaring (2.2) and using this gives

$$b(G)^2 \geq |H : H_\lambda|^2 b(H_\lambda)^2 \geq |H : H_\lambda|^2 |H_\lambda : \mathbf{F}(H_\lambda)| = \frac{|H : H_\lambda|}{|\mathbf{F}(H_\lambda)|} |H|.$$

Since $\mathbf{F}(H_\lambda)$ is nilpotent, $|\mathbf{F}(H_\lambda)| \leq \nu(H_\lambda)$, while (2.1) gives $|H : H_\lambda| \geq \nu(H_\lambda) \geq |\mathbf{F}(H_\lambda)|$. Hence $b(G)^2 \geq |H| = |G : \mathbf{F}(G)|$, contradicting the choice of G . $\square$

The reduction to irreducible modules is by induction, removing one irreducible summand at each step. The following lemma is the inductive step. The key point is that the ratio for the sum is bounded below by the product of the two ratios obtained from the summands.

Lemma 2.3. *Let H act faithfully and completely reducibly on $V = W \oplus U$, where W and U are nonzero H -submodules. Set $A = \mathbf{C}_H(W)$ and $\overline{H} = H/A$. If W contains an abundant vector for $\overline{H}$ and U contains an abundant vector for A , then V contains an abundant vector for H .*

Proof. The quotient $\overline{H}$ acts faithfully on W by the definition of A , and completely reducibly because W is an H -summand. The subgroup A is normal in H . Applying Clifford's theorem [16, Theorem 0.1] to each irreducible H -summand of V shows that the restriction of V to A is completely reducible, and hence so is its summand U . The action of A on U is faithful: if an element of A acts trivially on U , then it acts trivially on both W and U , and hence on V .

Fix $z = w + u$ with $w \in W$ and $u \in U$. Since the summands are H -invariant, $H_z = H_w \cap H_u$. Since $A \leq H_w$, we have $H_z A/A \leq (\overline{H})_w$ and $A \cap H_z = A_u$, whence

$$(2.3) \quad |H : H_z| = |H : H_z A| |H_z A : H_z| \geq |\overline{H} : (\overline{H})_w| |A : A_u|.$$

If $N \leq H_z$ is nilpotent, then both its image $NA/A \leq (\overline{H})_w$ and its intersection $N \cap A \leq A_u$ are nilpotent. Moreover, $|N| = |NA/A| |N \cap A|$. Therefore $|N| \leq \nu((\overline{H})_w) \nu(A_u)$, and maximising over N yields

$$(2.4) \quad \nu(H_z) \leq \nu((\overline{H})_w) \nu(A_u).$$

Now choose $w \in W$ with $\rho_{\overline{H}}(w) \geq 1$ and $u \in U$ with $\rho_A(u) \geq 1$, and set $z = w + u$. Combining (2.3) and (2.4) gives $\rho_H(z) \geq \rho_{\overline{H}}(w) \rho_A(u) \geq 1$, so z is abundant. $\square$

Corollary 2.4. *If Theorem 1.4 fails, it fails for some faithful irreducible $\mathbb{F}_p H$ -module.*

Proof. The zero module is not a counterexample: if it is faithful, then $H = 1$, and its only vector is abundant. Take a counterexample with $|V|$ minimal. By complete reducibility, V is a direct sum of irreducible elementary abelian submodules. If there is more than one summand, write $V = W \oplus U$ with W a single summand and U the sum of the others. Both W and U are smaller than V . As shown in the proof of Lemma 2.3, $H/\mathbf{C}_H(W)$ acts faithfully and completely reducibly on W , while $\mathbf{C}_H(W)$ acts faithfully and completely reducibly on U . Minimality therefore gives

an abundant vector for each action, and Lemma 2.3 gives one for V , a contradiction. Hence V is itself irreducible and elementary abelian, that is, a finite-dimensional vector space over a single prime field. $\square$

It remains to find an abundant vector in a faithful irreducible $\mathbb{F}_p H$ -module. We handle the primitive and imprimitive cases in Sections 6 and 8, after developing the tools they need.

3. COORDINATE CHAINS IN IMPRIMITIVE ACTIONS

The estimate in this section will be used in two settings. If $V = W_1 \oplus \cdots \oplus W_m$, then every vector $v \in V$ is uniquely determined by its components $v_i \in W_i$. If $\Omega = D_1 \dot{\cup} \cdots \dot{\cup} D_m$, then every subset $\Delta \subseteq \Omega$ is uniquely determined by the subsets $\Delta \cap D_i$. Thus in both settings the object whose stabiliser we study is represented by a tuple, while the group permutes the summands or parts and acts within them. We now formulate the estimate for abstract sets X_i . In the applications at the end of the section, X_i will be $\{i\} \times W_i$ in the linear case and $\{i\} \times \mathcal{P}(D_i)$ in the permutation case, where $\mathcal{P}(D_i)$ denotes the power set of D_i .

Let $\mathcal{X} = X_1 \dot{\cup} \cdots \dot{\cup} X_m$ be a finite set partitioned into nonempty parts, let $I = \{1, \dots, m\}$, and let $H \leq \text{Sym}(\mathcal{X})$ permute the parts transitively. Every $h \in H$ therefore induces a permutation of I . We write $j = i^h$ when $X_i^h = X_j$. We call i a *coordinate* and X_i its *coordinate set*.

The group H also acts on the Cartesian product $\prod_{i \in I} X_i$. If $x = (x_i)_{i \in I} \in \prod_{i \in I} X_i$, we call x a *coordinate tuple*. For $h \in H$ and $j = i^h$, the same element h sends $x_i \in X_i$ to $x_i^h \in X_j$, and its action on coordinate tuples is defined by $(x^h)_j = x_i^h$. The disjoint union $\mathcal{X}$ carries the given action of H , whereas the object whose stabiliser we study is the tuple $x \in \prod_i X_i$.

Here is a small example of the two actions. Take

$$X_1 = \{0_1, 1_1, 2_1\}, \quad X_2 = \{0_2, 1_2, 2_2\},$$

and let $H = \langle d, \tau \rangle$, where

$$d = (1_1 \ 2_1)(1_2 \ 2_2), \quad \tau = (0_1 \ 0_2)(1_1 \ 1_2)(2_1 \ 2_2).$$

Thus d acts inside the two coordinate sets and τ interchanges them. The disjoint union has six points, whereas $X_1 \times X_2$ has nine tuples. For instance,

$$(0_1, 1_2)^d = (0_1^d, 1_2^d) = (0_1, 2_2), \quad (0_1, 1_2)^\tau = (1_2^\tau, 0_1^\tau) = (1_1, 0_2).$$

In the calculation with τ , the entries of the image tuple are written in the order X_1, X_2 : the entry 1_2 is sent into X_1 , while 0_1 is sent into X_2 .

Let B be the kernel of the action of H on I , and write $P = H/B \leq \text{Sym}(I)$ for the permutation group induced by H on I . Thus B consists of the elements of H that stabilise every coordinate set setwise. We call P the *top group*.

Let M_i be the setwise stabiliser of X_i in H , and write

$$r_i : M_i \longrightarrow \text{Sym}(X_i), \quad r_i(h) = h|_{X_i}$$

for the restriction homomorphism. Define

$$J_i = r_i(M_i), \quad K_i = r_i(B).$$

We call J_i the *local group* on X_i . Since $B \trianglelefteq H$, we have $K_i \trianglelefteq J_i$. Although H is transitive on the family $\{X_1, \dots, X_m\}$, the group J_i need not be transitive on X_i .

Choose $t_i \in H$ with $X_1^{t_i} = X_i$, where $t_1 = 1$. The restriction of t_i is a bijection from X_1 to X_i . These bijections identify all the coordinate sets with X_1 . They also identify the setwise stabiliser in $\text{Sym}(\mathcal{X})$ of the partition $\{X_1, \dots, X_m\}$ with the wreath product $\text{Sym}(X_1) \wr S_m$ in its usual imprimitive action on $\mathcal{X}$. Hence H is a subgroup of this wreath product. The subgroup $\text{Sym}(X_1)^m$ consists of the permutations that stabilise each coordinate set setwise, and hence $B = H \cap \text{Sym}(X_1)^m$. This does not mean that the restrictions of elements of B to the individual

coordinate sets can be chosen independently. In the example, the two transpositions in d always occur together.

In the example above,

$$B = \langle d \rangle, \quad P = \langle \tau B \rangle \cong C_2, \quad J_1 = \langle (1_1 \ 2_1) \rangle,$$

and the H -orbits on $\mathcal{X}$ are

$$\mathcal{O}_0 = \{0_1, 0_2\} \quad \text{and} \quad \mathcal{O}_1 = \{1_1, 2_1, 1_2, 2_2\}.$$

For a coordinate tuple $x = (x_i)_{i \in I}$, define a map c_x on I by $c_x(i) = x_i^H$. Thus $c_x(i)$ is the H -orbit on $\mathcal{X}$ containing x_i . We call this orbit the *colour* of coordinate i in x . For a related use of inequivalent local colourings as colours in an imprimitive action, see [22, Section 2.3]. For $x = (0_1, 1_2)$, we have $c_x(1) = \mathcal{O}_0$ and $c_x(2) = \mathcal{O}_1$, so the two coordinates have different colours. For $y = (1_1, 2_2)$, both coordinates have colour $\mathcal{O}_1$.

The colour may also be read from the local action on X_1 . The chosen element t_i identifies X_i with X_1 by sending $u \in X_i$ to $u^{t_i^{-1}}$. Moreover,

$$x_i^H \cap X_1 = (x_i^{t_i^{-1}})^{J_1}.$$

Indeed, an element of H carries X_i to X_1 precisely when it has the form $t_i^{-1}m$ for some $m \in M_1$. In particular, the J_1 -orbit on the right is independent of the choice of t_i . Every H -orbit on $\mathcal{X}$ meets X_1 . Consequently, coordinates i and j have the same colour precisely when

$$c_x(i) = c_x(j) \iff (x_i^{t_i^{-1}})^{J_1} = (x_j^{t_j^{-1}})^{J_1}.$$

In the example, take $t_2 = \tau$. Then 0_1 and $1_2^{t_2^{-1}} = 1_1$ belong to the distinct J_1 -orbits $\{0_1\}$ and $\{1_1, 2_1\}$.

Since P acts on I , define the stabiliser in P of this colouring by

$$T_x = \{g \in P : c_x(i^g) = c_x(i) \text{ for every } i \in I\}.$$

An element of P lies in T_x precisely when it moves each index only to an index carrying the same colour. We call $|P : T_x|/\nu(T_x)$ the *top factor* associated with x . In the example, the nonidentity element of P interchanges the two indices. Consequently $T_x = 1$ for $x = (0_1, 1_2)$, while $T_y = P$ for $y = (1_1, 2_2)$.

To define the local factors, fix the order $1, \dots, m$ of the coordinate sets and define

$$B_0 = B, \quad B_i = B_{i-1} \cap \ker r_i, \quad L_i = r_i(B_{i-1}) \quad (1 \leq i \leq m).$$

Thus B_i is the pointwise stabiliser in B of $X_1 \cup \dots \cup X_i$, while L_i is the action on X_i induced by the elements of B that act trivially on $X_1, \dots, X_{i-1}$. Since $B \leq M_i$ and $\ker r_i \leq M_i$, induction gives $B_i \leq B$. Hence

$$L_i \leq K_i \leq J_i.$$

We call this the *coordinate chain* at i , and L_i its *bottom group*. Only L_i occurs in the inequality below. We retain K_i and J_i because the later arguments and finite calculations consider every chain $L \leq K \leq J$. The groups B_i and L_i may depend on the chosen order. For every order, however,

$$\prod_{i=1}^m |L_i| = |B|.$$

Indeed, $|L_i| = |B_{i-1} : B_i|$, and the faithful action of H on $\mathcal{X}$ gives $B_m = 1$.

The same example shows why these chains are needed. The element d acts nontrivially on both X_1 and X_2 , but its action on either set determines its action on the other. With the order $1, 2$, we have

$$B_0 = B, \quad B_1 = 1, \quad L_1 \cong C_2, \quad L_2 = 1.$$

After the action on X_1 has been taken into account, no nonidentity element of B remains to act on X_2 . For comparison, if

$$B = \langle (1_1 \ 2_1), (1_2 \ 2_2) \rangle,$$

then the actions on X_1 and X_2 are independent, and $L_1 \cong L_2 \cong C_2$.

We shall also use the following elementary inequality. If $S \leq M \leq L$, then $\nu(S) \leq \nu(M)$ and

$$(3.1) \quad \frac{|L : S|}{\nu(S)} \geq \frac{|L : M|}{\nu(M)}.$$

Theorem 3.1 (Coordinate chain inequality). *For the coordinate chains $L_i \trianglelefteq K_i \trianglelefteq J_i$ associated with the chosen order, every coordinate tuple $x = (x_i)_{i \in I}$ satisfies*

$$(3.2) \quad \rho_H(x) \geq \frac{|P : T_x|}{\nu(T_x)} \prod_{i=1}^m \rho_{L_i}(x_i).$$

Proof. We first separate the action of the stabiliser on the coordinate sets from its intersection with B . Define

$$B_x = B \cap H_x, \quad R = H_x B / B \leq P.$$

Then $|H : H_x| = |P : R| |B : B_x|$. If $N \leq H_x$ is nilpotent, then the homomorphism from N to R has kernel $N \cap B = N \cap B_x$. Its image and kernel are nilpotent, and hence $|N| \leq \nu(R)\nu(B_x)$. Therefore

$$(3.3) \quad \rho_H(x) \geq \frac{|P : R|}{\nu(R)} \frac{|B : B_x|}{\nu(B_x)}.$$

We next bound the top contribution in (3.3). Suppose $h \in H_x$ carries X_i to X_j . Since $x^h = x$, the element h also carries x_i to x_j . These two entries therefore have the same colour. It follows that the image R of H_x in P preserves the map c_x , so $R \leq T_x$. Applying (3.1) gives

$$(3.4) \quad \frac{|P : R|}{\nu(R)} \geq \frac{|P : T_x|}{\nu(T_x)}.$$

It remains to bound the contribution from B . Recall that $B_x = B \cap H_x$. If $b \in B_x$, then $i^b = i$ because $b \in B$, while $x^b = x$ because $b \in H_x$. Hence $x_i^b = (x^b)_i = x_i$ for every i . Thus B_x fixes every entry of x . This need not be true of the whole stabiliser H_x : in the example above, $\tau \in H_{(0_1, 0_2)} \setminus B$ fixes the tuple $(0_1, 0_2)$, although $0_1^\tau = 0_2$ and $0_2^\tau = 0_1$.

For $1 \leq i \leq m$, define $S_i = r_i(B_x \cap B_{i-1})$. Since $B_x \cap B_{i-1} \leq B_{i-1}$, we have $S_i \leq L_i$. If $s \in S_i$, then $s = r_i(b)$ for some $b \in B_x \cap B_{i-1}$, and the preceding paragraph gives $x_i^s = x_i^b = x_i$. Hence $S_i \leq (L_i)_{x_i}$. This inclusion may be strict: an element of L_i fixing x_i need not have a preimage in B_{i-1} that fixes $x_{i+1}, \dots, x_m$.

Recall that $B_m = 1$. The normality of the groups B_i in B gives a chain

$$B = B_x B_0 \geq B_x B_1 \geq \dots \geq B_x B_m = B_x.$$

Multiplying the successive indices in this chain gives

$$|B : B_x| = \prod_{i=1}^m |B_x B_{i-1} : B_x B_i|.$$

For each i , the product formula gives the first equality below. Since $B_i \leq B_{i-1}$, Dedekind's modular law gives the second:

$$|B_x B_{i-1} : B_x B_i| = |B_{i-1} : B_{i-1} \cap (B_x B_i)| = |B_{i-1} : (B_x \cap B_{i-1}) B_i|.$$

By definition, $r_i(B_{i-1}) = L_i$, and the kernel of r_i on B_{i-1} is $B_{i-1} \cap \ker r_i = B_i$. Since $r_i(B_i) = 1$, the restriction $r_i|_{B_{i-1}}$ maps $(B_x \cap B_{i-1}) B_i$ onto S_i . A surjective homomorphism preserves the index of a subgroup containing its kernel, so $|B_{i-1} : (B_x \cap B_{i-1}) B_i| = |L_i : S_i|$. Multiplying these equalities over all i gives

$$(3.5) \quad |B : B_x| = \prod_{i=1}^m |L_i : S_i|.$$

Let $N \leq B_x$ be nilpotent. Since $B_i \trianglelefteq B$, the groups $N \cap B_i$ form a normal series from N to 1. For each i , the kernel of $r_i|_{N \cap B_{i-1}}$ is $N \cap B_i$. The first isomorphism theorem therefore gives

$$\frac{N \cap B_{i-1}}{N \cap B_i} \cong r_i(N \cap B_{i-1}) \leq S_i.$$

The image is nilpotent, so each factor has order at most $\nu(S_i)$, and hence

$$|N| \leq \prod_{i=1}^m \nu(S_i).$$

Maximising over all nilpotent subgroups N of B_x yields

$$(3.6) \quad \nu(B_x) \leq \prod_{i=1}^m \nu(S_i).$$

Combining (3.5) and (3.6), and then applying (3.1) with $S = S_i$, $M = (L_i)_{x_i}$ and $L = L_i$, gives

$$(3.7) \quad \frac{|B : B_x|}{\nu(B_x)} \geq \prod_{i=1}^m \frac{|L_i : S_i|}{\nu(S_i)} \geq \prod_{i=1}^m \rho_{L_i}(x_i).$$

The theorem now follows from (3.3), (3.4) and (3.7). $\square$

We now apply Theorem 3.1 in the two settings described at the beginning of the section. The tags below make the coordinate sets disjoint.

For the linear application, suppose that H acts faithfully and imprimitively on $V = W_1 \oplus \cdots \oplus W_m$, permuting the summands transitively. The summands are not disjoint as sets because they all contain the zero vector. We therefore use the tagged copies $X_i = \{i\} \times W_i$. The tag i serves only to distinguish these copies. The action on them is $(i, v)^h = (i^h, v^h)$. The map

$$\prod_{i \in I} X_i \longrightarrow V, \quad ((i, v_i))_{i \in I} \longmapsto \sum_{i \in I} v_i$$

is a bijection because the sum is direct. If $x = ((i, v_i))_{i \in I}$ and $j = i^h$, then the j th entry of x^h is (j, v_i^h) . Hence the vector corresponding to x^h is

$$\sum_{i \in I} v_i^h = \left(\sum_{i \in I} v_i \right)^h,$$

so the bijection is H -equivariant. Thus the coordinate tuple corresponds to the coordinate decomposition of a vector, and the stabiliser of the tuple is the stabiliser of that vector. The tagged action is faithful, and we identify the local permutation group on X_i with the corresponding group induced on W_i .

The action on $\mathcal{X}$ is not transitive: the tagged zero vectors form one orbit, while the nonzero vectors need not form a single orbit. Since Theorem 1.5 concerns transitive actions, it cannot be applied directly to $\mathcal{X}$.

Nor would an abundant subset $\Delta \subseteq \mathcal{X}$ by itself give an abundant vector. Such a subset may contain several elements of one coordinate set X_i and none of another, whereas a coordinate tuple contains exactly one element of each X_i . The map

$$\mathcal{P}(\mathcal{X}) \longrightarrow V, \quad \Delta \longmapsto \sum_{(i, w) \in \Delta} w$$

is H -equivariant. If this map sends Δ to v , then $H_\Delta \leq H_v$, and this inclusion may be strict. It follows from (3.1) that $\rho_H(\Delta) \geq \rho_H(v)$. Thus the abundance of Δ does not imply that v is abundant.

Section 8 proves the imprimitive linear case by applying Theorem 1.5, or its weighted strengthening Theorem 5.1, to the top group $P = H/B$ acting on I . Theorem 3.1 then combines the resulting top estimate with the local ratios $\rho_{L_i}(x_i)$ arising from the bottom groups L_i .

For the application to permutation groups, suppose that H acts faithfully on $\Omega = D_1 \dot{\cup} \dots \dot{\cup} D_m$ and permutes the sets D_i transitively. Take $X_i = \{i\} \times \mathcal{P}(D_i)$ with the natural action. The map

$$\prod_{i \in I} X_i \longrightarrow \mathcal{P}(\Omega), \quad ((i, \Delta_i))_{i \in I} \longmapsto \bigcup_{i \in I} \Delta_i$$

is a bijection because the sets D_i are disjoint. If $x = ((i, \Delta_i))_{i \in I}$, then h carries Δ_i to $\Delta_i^h \subseteq D_{i^h}$, and the subset corresponding to x^h is

$$\bigcup_{i \in I} \Delta_i^h = \left(\bigcup_{i \in I} \Delta_i \right)^h.$$

Thus the bijection is H -equivariant, and the stabiliser of the coordinate tuple is the setwise stabiliser of the corresponding subset. For a subset $\Delta \subseteq \Omega$, the local subset at i is $\Delta_i = \Delta \cap D_i$. The tagged action is faithful, since its kernel fixes every singleton of Ω . The groups induced by M_i on $\mathcal{P}(D_i)$ and D_i are naturally isomorphic, and we identify them. The bijections $X_1 \longrightarrow X_i$ induced by the elements t_i identify the local groups and their orbits. When a local orbit on X_1 is used on X_i , we denote its image by the same symbol.

4. ABUNDANT SUBSETS OF SOLVABLE TRANSITIVE GROUPS

We now prove the first of the two results about permutation groups promised in the introduction: a solvable transitive group always has an abundant subset.

The induction uses an *unrefinable* block system, by which we mean a nontrivial block system with no strictly finer nontrivial block system. Its local group is primitive, since a system of imprimitivity for the local action would transport to a proper refinement of the original block system. Here the *local group* is the permutation group induced on a block by its setwise stabiliser, and the quotient $\bar{P}$ is the permutation group induced by P on the set of blocks.

Definition 4.1. Let D be a finite set and let $J \leq \text{Sym}(D)$ be a primitive permutation group. A J -orbit $\mathcal{O}$ on the power set $\mathcal{P}(D)$ is a *chain-abundant orbit* if, for every chain of subgroups $L \leq K \leq J$, there exists $\Delta \in \mathcal{O}$ with $\rho_L(\Delta) \geq 1$. The same orbit $\mathcal{O}$ is used for every chain, but the representative chosen from it may vary.

Recall that the degree of a finite solvable primitive permutation group is a prime power. The possible nontrivial degrees at most nine are therefore 2, 3, 4, 5, 7, 8 and 9. We deal with the smallest degrees here and use a short computation for the rest later.

Lemma 4.2. *Let J be a solvable primitive permutation group on a finite set D with $|D| = 3$ or 4. Then the singletons in D and the subsets of D of size 2 form two distinct chain-abundant J -orbits on $\mathcal{P}(D)$.*

Proof. Fix a chain $L \leq K \leq J$. If $|D| = 3$, then the primitive groups are C_3 and S_3 , and L is one of 1, C_3 , S_3 . For $L = 1$ both ratios are 1. For $L = C_3$, both a singleton and a subset of size 2 have trivial stabiliser, so their abundance ratios are 3. For $L = S_3$, both stabilisers have order 2, so the abundance ratios are 3/2.

If $|D| = 4$, then the primitive groups are A_4 and S_4 , and the possible bottom groups are 1, C_2 , V_4 , A_4 , S_4 . For each L , the largest ratios among the singletons and among the subsets of size 2 are

L	1	C_2	V_4	A_4	S_4
singleton	1	2	4	4/3	4/3
size 2	1	2	1	3	3/2

Here each maximum is taken over the relevant J -orbit. For example, with $L = \langle (12)(34) \rangle \cong C_2$ the subset $\{1, 3\}$ has trivial stabiliser in L and ratio 2, whereas $\{1, 2\}$ gives only 1/2. One checks the entries from the orbit sizes and from the orders of the relevant setwise stabilisers, $\nu(C_2) = 2$, $\nu(C_3) = \nu(S_3) = 3$, $\nu(V_4) = \nu(A_4) = 4$ and $\nu(S_4) = 8$. Every entry is at least 1. $\square$

Lemma 4.3. *Let J be a solvable primitive permutation group on a finite set D with $|D| \in \{5, 7, 8, 9\}$. Then J has at least two distinct chain-abundant orbits on $\mathcal{P}(D)$. The two orbits may be chosen on subsets of sizes 1 and 2, except when J is $\text{AGL}(1, 9)$ or $\text{AGL}(2, 3)$ in degree nine. For these two groups, subsets of sizes 2 and 3 suffice.*

Proof. Among the 21 solvable primitive groups of degree at most nine in the primitive groups library of GAP [7], exactly 16 have degree in $\{5, 7, 8, 9\}$. Up to relabelling the elements of D , the group J is one of these 16 groups. Relabelling does not change whether an orbit is chain-abundant. The calculation in Appendix A.2 checks all 21 groups and hence, in particular, the 16 cases relevant here. For each relevant group J , it considers every subgroup L that can occur as the bottom group of a chain $L \trianglelefteq K \trianglelefteq J$ and every J -orbit $\mathcal{O}$ on $\mathcal{P}(D)$. It verifies that two orbits of the sizes stated in the lemma contain, for every such L , a subset Δ satisfying $\rho_L(\Delta) \geq 1$. For the two exceptional groups the point stabilisers $\Gamma\text{L}(1, 9)$ and $\text{GL}(2, 3)$ contain nilpotent subgroups of order 16. Thus every singleton Δ satisfies $\rho_J(\Delta) \leq 9/16 < 1$ for the chain $L = K = J$, so no singleton orbit is chain-abundant. For each of these two groups, Appendix A.2 finds two chain-abundant J -orbits, one consisting of subsets of size 2 and one consisting of subsets of size 3. $\square$

For larger degrees we use Gluck's theorem on regular subsets in place of any computation.

Lemma 4.4. *Let J be a solvable primitive permutation group on a finite set D with $|D| > 9$. Then there exists a regular subset $\Delta_0 \subseteq D$ with $|\Delta_0| \neq |D|/2$. In particular, the J -orbits of Δ_0 and Δ_0^c are distinct and chain-abundant.*

Proof. By [9, Theorem 1] there is a regular subset Δ_0 with $|\Delta_0| \neq |D|/2$. Its complement Δ_0^c also has trivial stabiliser. The subsets Δ_0 and Δ_0^c have different sizes and therefore lie in distinct J -orbits. For every chain $L \trianglelefteq K \trianglelefteq J$, both L_{Δ_0} and $L_{\Delta_0^c}$ are trivial. Hence $\rho_L(\Delta_0) = \rho_L(\Delta_0^c) = |L| \geq 1$. $\square$

The next observation concerns a coordinate tuple with exactly two colours.

Lemma 4.5. *Let $x = (x_i)_{i \in I}$ be a coordinate tuple with exactly two colours. Let $\mathcal{O}$ be one of these colours and set $\Gamma = \{i \in I : c_x(i) = \mathcal{O}\}$. Then $T_x \leq P_\Gamma$.*

Proof. Let $g \in T_x$. If $i \in \Gamma$, then $c_x(i^g) = c_x(i) = \mathcal{O}$, so $i^g \in \Gamma$. Thus g stabilises Γ setwise and hence belongs to P_Γ . $\square$

The case where the blocks D_i have size two needs a different argument.

Lemma 4.6. *Let $I = \{1, \dots, m\}$. For each $i \in I$, let D_i be a set of size 2, let $L_i \leq \text{Sym}(D_i)$, and choose a singleton $E_i \subseteq D_i$. Fix $\Gamma \subseteq I$, and define*

$$\Delta_i = \begin{cases} E_i, & i \in \Gamma, \\ \emptyset, & i \notin \Gamma, \end{cases} \quad \Delta'_i = \begin{cases} \emptyset, & i \in \Gamma, \\ E_i, & i \notin \Gamma. \end{cases}$$

Then

$$\left(\prod_{i \in I} \rho_{L_i}(\Delta_i) \right) \left(\prod_{i \in I} \rho_{L_i}(\Delta'_i) \right) = 1.$$

Consequently at least one of the two products is at least 1.

Proof. Since D_i has two elements, either $L_i = 1$ or $L_i = \text{Sym}(D_i) \cong C_2$. If $L_i = 1$, both local abundance ratios are 1. If $L_i \cong C_2$, the singleton has abundance ratio 2, while the empty set has abundance ratio 1/2. Thus

$$\rho_{L_i}(\Delta_i) \rho_{L_i}(\Delta'_i) = 1$$

for every $i \in I$. Multiplying these equalities gives the displayed identity. Since both products are positive, at least one of them is at least 1. $\square$

Proof of Theorem 1.5. Induct on $|\Omega|$, the case $|\Omega| = 1$ being trivial. Suppose first that P is primitive. If $|\Omega| > 9$, then a regular subset supplied by Lemma 4.4 is abundant. Degrees three and four are covered by Lemma 4.2, degrees 5, 7, 8, 9 by Lemma 4.3, and in degree two a singleton has trivial stabiliser. These are all primitive cases.

Now let P be imprimitive, and fix an unrefinable block system $D_1, \dots, D_m$. Let $I = \{1, \dots, m\}$, where i indexes D_i , and let J be the local primitive group on D_1 . Let $\bar{P} \leq \text{Sym}(I)$ be the group induced by P on I . By induction choose $\Gamma \subseteq I$ with $\rho_{\bar{P}}(\Gamma) \geq 1$. Since $\bar{P}$ is nontrivial, $\rho_{\bar{P}}(\emptyset) = \rho_{\bar{P}}(I) = 1/\nu(\bar{P}) < 1$. The abundant subset Γ is therefore neither empty nor all of I . Thus both Γ and $I \setminus \Gamma$ are nonempty.

If $J \not\cong C_2$, the construction below adapts the proof of [9, Theorem 2], with an abundant top subset and two chain-abundant local orbits in place of the corresponding regular subsets. The group J has at least two distinct chain-abundant orbits by Lemmas 4.2, 4.3 and 4.4. For each $i \in I$, choose $\Delta_i \subseteq D_i$ in the first orbit if $i \in \Gamma$, and in the second orbit otherwise, so that $\rho_{L_i}(\Delta_i) \geq 1$. Let Δ be the union of the subsets Δ_i . Let $T \leq \bar{P}$ be the subgroup preserving the colours of the corresponding coordinate tuple. By Lemma 4.5, $T \leq \bar{P}_\Gamma$. Hence $|\bar{P} : T|/\nu(T) \geq \rho_{\bar{P}}(\Gamma) \geq 1$. Every local factor in (3.2) is also at least 1. It follows that $\rho_P(\Delta) \geq 1$.

If $J \cong C_2$, choose a singleton $E_i \subseteq D_i$ for every $i \in I$. Use E_i when $i \in \Gamma$ and the empty subset of D_i otherwise. The two local orbits are distinct. If $T \leq \bar{P}$ is the subgroup preserving the colours of the corresponding coordinate tuple, then $T \leq \bar{P}_\Gamma = \bar{P}_{\Gamma^c}$, and the top factor is at least 1. By Lemma 4.6, either this choice of local subsets or the choice obtained by interchanging the singleton and the empty set on every block has product of local ratios at least 1. Let T' be the subgroup preserving the colours of the interchanged tuple. Its singleton coordinates are indexed by Γ^c , so $T' \leq \bar{P}_{\Gamma^c} = \bar{P}_\Gamma$. Thus the top factor is at least 1 for either tuple. Let Δ be the union of the local subsets in whichever of the two choices has product of local ratios at least 1. Applying Theorem 3.1 to the corresponding coordinate tuple gives $\rho_P(\Delta) \geq 1$. $\square$

5. A WEIGHTED TRANSITIVE THEOREM

Theorem 5.1 is a weighted version of Theorem 1.5. Each point $i \in \Omega$ carries two positive real weights z_i and n_i , with the only condition that $z_i n_i^3 \geq 1$. We call this inequality the *cubic condition*. For a subset $\Delta \subseteq \Omega$, the weight z_i is used when $i \in \Delta$, and n_i is used when $i \notin \Delta$. The theorem asserts that Δ can be chosen so that the resulting weighted abundance ratio is at least 1. In the cubic case of Section 8, the theorem is applied to the permutation group induced on the index set I of the summands of an imprimitive module.

Theorem 5.1. *Let P be a finite solvable transitive permutation group on Ω . For each $i \in \Omega$, let z_i, n_i be positive real numbers satisfying $z_i n_i^3 \geq 1$. Then there exists $\Delta \subseteq \Omega$ such that*

$$\rho_P(\Delta) \prod_{i \in \Delta} z_i \prod_{i \notin \Delta} n_i \geq 1.$$

Before proving the theorem, we establish several lemmas.

Lemma 5.2. *Let P be a finite transitive permutation group on Ω . Suppose that*

$$(5.1) \quad \max_{\Delta \subseteq \Omega} \rho_P(\Delta) r^{|\Omega| - 4|\Delta|} \geq 1$$

for every $r > 0$. For each $i \in \Omega$, let z_i, n_i be positive real numbers satisfying $z_i n_i^3 \geq 1$. Then there exists $\Delta \subseteq \Omega$ such that

$$\rho_P(\Delta) \prod_{i \in \Delta} z_i \prod_{i \notin \Delta} n_i \geq 1.$$

Proof. Write $d = |\Omega|$. Since $z_i \geq n_i^{-3}$ for every $i \in \Omega$, it is enough to prove the conclusion when $z_i = n_i^{-3}$ for every $i \in \Omega$. Define

$$r = \left(\prod_{i \in \Omega} n_i \right)^{1/d}.$$

By (5.1), choose $\Delta \subseteq \Omega$ such that $\rho_P(\Delta)r^{d-4|\Delta|} \geq 1$. The setwise stabilisers of Δ and Δ^g are conjugate for every $g \in P$, so $\rho_P(\Delta^g) = \rho_P(\Delta)$. Since P is transitive, the number of elements $g \in P$ such that $i \in \Delta^g$ is the same for every $i \in \Omega$. Counting the pairs $(i, g) \in \Omega \times P$ with $i \in \Delta^g$ shows that this number is $|P||\Delta|/d$. Consequently

$$\prod_{g \in P} \left(\rho_P(\Delta^g) \prod_{i \in \Delta^g} n_i^{-3} \prod_{i \notin \Delta^g} n_i \right) = \rho_P(\Delta)^{|P||\Delta|/d} r^{|P|(d-4|\Delta|)} = \left(\rho_P(\Delta)r^{d-4|\Delta|} \right)^{|P|} \geq 1.$$

All the factors are positive. Since their product is at least 1, choose $g \in P$ for which the factor indexed by g is at least 1. The inequalities $z_i \geq n_i^{-3}$ then show that Δ^g satisfies the conclusion. $\square$

We next formulate the local condition used to prove (5.1) for $r \geq 1$.

Definition 5.3. Let J be a primitive permutation group on a finite set D . Let $\mathcal{Z}$ and $\mathcal{N}$ be nonempty disjoint collections of J -orbits on the power set $\mathcal{P}(D)$. For each chain $L \trianglelefteq K \trianglelefteq J$ and each $r > 0$, define

$$Z_L(r) = \max_{\substack{\Delta \subseteq D \\ \Delta^J \in \mathcal{Z}}} \rho_L(\Delta) r^{|D|-4|\Delta|}, \quad N_L(r) = \max_{\substack{\Delta \subseteq D \\ \Delta^J \in \mathcal{N}}} \rho_L(\Delta) r^{|D|-4|\Delta|}.$$

We call the ordered pair $(\mathcal{Z}, \mathcal{N})$ a *cubic pair* for J at r if $Z_L(r)N_L(r)^3 \geq 1$ for every such chain.

Lemma 5.4. Let P be a finite transitive permutation group on Ω . Let $D_1, \dots, D_m$ be an unrefinable block system. Set $I = \{1, \dots, m\}$, where i indexes D_i , and let $\bar{P} \leq \text{Sym}(I)$ be the group induced by P on I . Let J be the local group on D_1 , let $r > 0$, and suppose that $(\mathcal{Z}, \mathcal{N})$ is a cubic pair for J at r . Suppose that, for every choice of positive real numbers z_i, n_i , $i \in I$, satisfying $z_i n_i^3 \geq 1$ for every $i \in I$, there exists $\Gamma \subseteq I$ such that

$$(5.2) \quad \rho_{\bar{P}}(\Gamma) \prod_{i \in \Gamma} z_i \prod_{i \notin \Gamma} n_i \geq 1.$$

Then there exists $\Delta \subseteq \Omega$ such that $\rho_P(\Delta)r^{|\Omega|-4|\Delta|} \geq 1$.

Proof. For each $i \in I$, let $L_i \trianglelefteq K_i \trianglelefteq J_i$ be the coordinate chain at i . The local identifications in Section 3 allow us to use $\mathcal{Z}$ and $\mathcal{N}$ on each D_i . Set $z_i = Z_{L_i}(r)$ and $n_i = N_{L_i}(r)$. The cubic pair condition gives $z_i n_i^3 \geq 1$, so the hypothesis on $\bar{P}$ supplies $\Gamma \subseteq I$ satisfying (5.2). For each $i \in I$, choose a subset of D_i attaining $Z_{L_i}(r)$ if $i \in \Gamma$, and one attaining $N_{L_i}(r)$ otherwise. Let Δ be their union, and let $T \leq \bar{P}$ be the subgroup preserving the colours of the corresponding coordinate tuple. If $\Gamma = \emptyset$ or $\Gamma = I$, then $T \leq \bar{P}_\Gamma$ trivially. Otherwise the two orbit families are disjoint, and again $T \leq \bar{P}_\Gamma$. In either case, $|\bar{P} : T|/\nu(T) \geq \rho_{\bar{P}}(\Gamma)$. Multiplying (3.2) by $r^{|\Omega|-4|\Delta|}$ and grouping the weight block by block gives

$$\rho_P(\Delta)r^{|\Omega|-4|\Delta|} \geq \rho_{\bar{P}}(\Gamma) \prod_{i \in \Gamma} z_i \prod_{i \notin \Gamma} n_i \geq 1,$$

which proves the assertion. $\square$

We next construct cubic pairs for the primitive groups other than C_2 . We begin with the case in which a regular subset is available.

Lemma 5.5. Let J be a nontrivial primitive permutation group on a finite set D . Let $\Delta_0 \subseteq D$ be a regular subset with $|\Delta_0| = a \neq |D|/2$. Then $\mathcal{Z} = \{(\Delta_0^c)^J\}$ and $\mathcal{N} = \{\Delta_0^J, \{\emptyset\}\}$ form a cubic pair for J at every $r > 0$.

Proof. The three orbits are distinct. Fix a chain $L \trianglelefteq K \trianglelefteq J$. Regularity gives $\rho_L(\Delta_0) = \rho_L(\Delta_0^c) = |L|$, while $\rho_L(\emptyset) = \nu(L)^{-1}$. With $s = r^{4a}$,

$$Z_L(r) = |L| r^{4a-3|D|}, \quad N_L(r) = \max\{|L| r^{|D|-4a}, \nu(L)^{-1} r^{|D|}\},$$

so

$$Z_L(r)N_L(r)^3 = \max\left\{\frac{|L|^4}{s^2}, \frac{|L|}{\nu(L)^3} s\right\}.$$

If $s \leq |L|\nu(L)$, the first term is at least $(|L|/\nu(L))^2$. If $s \geq |L|\nu(L)$, the second term is at least $(|L|/\nu(L))^2$. As $\nu(L) \leq |L|$, the product is at least 1. $\square$

Lemma 5.6. *Let J be a solvable primitive permutation group on a finite set D with $|D| \in \{3, 4, 5, 7, 8, 9\}$. Then J admits a cubic pair for every $r \geq 1$. The following choices suffice. When $|D| = 3$ or 4 , let $\mathcal{O}_k$ denote the J -orbit consisting of the subsets of D of size k .*

- (1) *If $|D| = 3$, then $J \cong C_3$ or S_3 . Take $\mathcal{Z} = \{\mathcal{O}_0\}$ and $\mathcal{N} = \{\mathcal{O}_1\}$.*
- (2) *If $|D| = 4$, then $J \cong A_4$ or S_4 . Take $\mathcal{Z} = \{\mathcal{O}_0, \mathcal{O}_2\}$ and $\mathcal{N} = \{\mathcal{O}_1\}$.*
- (3) *If $|D| \in \{5, 7, 8, 9\}$, let $\mathcal{A}$ and $\mathcal{B}$ be distinct chain-abundant J -orbits on $\mathcal{P}(D)$ such that the subsets in $\mathcal{A}$ and $\mathcal{B}$ have sizes a_Z and a_N , respectively, and $a_Z + 3a_N \leq |D|$. Such orbits exist by Lemma 4.3. Set $\mathcal{Z} = \{\mathcal{A}\}$ and $\mathcal{N} = \{\mathcal{B}\}$.*

Proof. In (1), the powers of r occurring in $Z_L(r)$ and $N_L(r)$ are r^3 and r^{-1} , respectively. They therefore cancel in $Z_L(r)N_L(r)^3$. The value is 1, 9, 9/8 for $L = 1, C_3, S_3$ respectively. In (2), set $\alpha = \max_{|\Delta|=2} \rho_L(\Delta)$ and $\beta = \max_{|\Delta|=1} \rho_L(\Delta)$. With $s = r^4$, one has $N_L(r) = \beta$ and $Z_L(r) = \max\{\alpha/s, s/\nu(L)\} \geq \sqrt{\alpha/\nu(L)}$. Hence $Z_L(r)N_L(r)^3 \geq \beta^3 \sqrt{\alpha/\nu(L)}$, and it suffices that $\alpha\beta^6 \geq \nu(L)$. The five possible bottom groups give

L	1	C_2	V_4	A_4	S_4
$\alpha\beta^6$	1	128	4096	4096/243	2048/243
$\nu(L)$	1	2	4	4	8

and $\alpha\beta^6 \geq \nu(L)$ in every column. In (3), both orbits are chain-abundant. Hence each ρ_L may be taken at least 1, and then $Z_L(r)N_L(r)^3 \geq r^{(|D|-4a_Z)+3(|D|-4a_N)} = r^{4(|D|-a_Z-3a_N)} \geq 1$ for $r \geq 1$, the sizes being $(a_Z, a_N) = (2, 1)$ or $(3, 2)$ as in Lemma 4.3. $\square$

Lemma 5.7. *Let P be a finite transitive permutation group on Ω . Let $D_1, \dots, D_m$ be a block system with $|D_i| = 2$ for every i , and let $I = \{1, \dots, m\}$, where i indexes D_i . Suppose that the local group on D_1 is isomorphic to C_2 , and let $\bar{P} \leq \text{Sym}(I)$ be the group induced by P on I . If $\bar{P}$ has an abundant subset, then, for every $r > 0$, there exists $\Delta \subseteq \Omega$ such that*

$$\rho_P(\Delta) r^{|\Omega|-4|\Delta|} \geq 1.$$

Proof. Choose $\Gamma \subseteq I$ with $\rho_{\bar{P}}(\Gamma) \geq 1$. For each $i \in I$, let L_i be the bottom group at i and choose a singleton $E_i \subseteq D_i$. Set

$$\Delta = \bigcup_{i \in \Gamma} E_i, \quad \Delta' = \bigcup_{i \notin \Gamma} E_i.$$

Let $T, T' \leq \bar{P}$ be the subgroups preserving the colours of the coordinate tuples corresponding to Δ and Δ' , respectively. In the first tuple, the singleton entries are indexed by Γ , while in the second they are indexed by $I \setminus \Gamma$. Thus $T, T' \leq \bar{P}_\Gamma$. By (3.1),

$$\frac{|\bar{P} : T|}{\nu(T)} \geq \rho_{\bar{P}}(\Gamma), \quad \frac{|\bar{P} : T'|}{\nu(T')} \geq \rho_{\bar{P}}(\Gamma),$$

so both top factors are at least 1.

Lemma 4.6 gives

$$\left(\prod_{i \in I} \rho_{L_i}(\Delta \cap D_i) \right) \left(\prod_{i \in I} \rho_{L_i}(\Delta' \cap D_i) \right) = 1.$$

Moreover, $|\Omega| = 2m$, $|\Delta| = |\Gamma|$ and $|\Delta'| = m - |\Gamma|$. Hence the corresponding weight factors are

$$r^{2m-4|\Gamma|} \quad \text{and} \quad r^{2m-4(m-|\Gamma|)} = r^{-(2m-4|\Gamma|)},$$

which are reciprocal. Combining each local product with its weight factor therefore gives two positive numbers whose product is 1.

Apply Theorem 3.1 to the two coordinate tuples and multiply the resulting inequalities by their respective weight factors. Consequently, the asserted inequality holds for at least one of Δ and Δ' . $\square$

Proof of Theorem 5.1. We prove the theorem by induction on $d = |\Omega|$. If $d = 1$, write $\Omega = \{i\}$. The group P is trivial. If $n_i \geq 1$, then $\Delta = \emptyset$ gives the required inequality. If $n_i < 1$, then $z_i \geq n_i^{-3} > 1$, and $\Delta = \Omega$ gives the required inequality.

Assume that $d \geq 2$ and that the theorem holds for smaller degrees. By Lemma 5.2, it suffices to prove (5.1) for every $r > 0$. Suppose first that $0 < r \leq 1$. By Theorem 1.5, choose $\Delta \subseteq \Omega$ with $\rho_P(\Delta) \geq 1$. Since $P_\Delta = P_{\Delta^c}$, we have $\rho_P(\Delta) = \rho_P(\Delta^c)$. Replacing Δ by its complement if necessary, we may assume that $|\Delta| \geq d/2$. It follows that $d - 4|\Delta| \leq -d \leq 0$, and hence $\rho_P(\Delta)r^{d-4|\Delta|} \geq 1$.

Now fix $r \geq 1$. Every solvable primitive permutation group of degree at least three has a cubic pair at r . For degree greater than nine, use the regular subset supplied by Lemma 4.4 and apply Lemma 5.5. In degrees 3, 4, 5, 7, 8, 9, such a pair is supplied by Lemma 5.6.

Suppose first that P is primitive. If $P \cong C_2$, take $D_1 = \Omega$. The local group is $P \cong C_2$, and the group $\bar{P}$ induced on $I = \{1\}$ is trivial, so it has an abundant subset. Hence Lemma 5.7 applies. Otherwise choose a cubic pair for P at r as above. Taking $L = K = P$ in its defining inequality gives $Z_P(r)N_P(r)^3 \geq 1$. Hence $\max\{Z_P(r), N_P(r)\} \geq 1$. Choose a subset Δ attaining whichever of $Z_P(r)$ and $N_P(r)$ is larger. Then $\rho_P(\Delta)r^{d-4|\Delta|} \geq 1$, so (5.1) holds.

Now suppose that P is imprimitive. Choose an unrefinable block system and number its blocks D_i , where $I = \{1, \dots, m\}$ is the index set. Let J be the local group on D_1 , and let $\bar{P} \leq \text{Sym}(I)$ be the group induced by P on I . If $J \cong C_2$, Theorem 1.5 provides an abundant subset of I for $\bar{P}$, so Lemma 5.7 applies. Otherwise J has a cubic pair at r by the preceding paragraph. The group $\bar{P}$ is solvable and transitive of degree $m < d$, so the induction hypothesis gives the weighted hypothesis of Lemma 5.4. That lemma gives $\Delta \subseteq \Omega$ with $\rho_P(\Delta)r^{d-4|\Delta|} \geq 1$.

Together with the case $0 < r \leq 1$, this proves (5.1) for every $r > 0$. Lemma 5.2 completes the proof. $\square$

6. PRIMITIVE LINEAR GROUPS

We now prove the abundance theorem for faithful irreducible primitive modules. The non-metacyclic case follows from the classification of actions without a regular orbit [12] and the finite verification in Appendix A. The remaining metacyclic case is handled by a semilinear argument.

6.1. The nonmetacyclic case. We require only the primitive part of the classification of residual actions.

Theorem 6.1 (Primitive residual actions). *Let H be a finite solvable group acting faithfully, irreducibly and primitively on a finite $\mathbb{F}_p$ -space V . If H is not metacyclic and has no regular orbit, then its action is linearly equivalent to one represented in the linearly primitive collection described in Appendix A.*

Proof. Every primitive irreducible linear action is quasi-primitive [16, p. 27], so the classification of residual actions applies. Holt and Yang [12] gave the original classification and accompanying matrix data. The corrected Holt–Yang data are proved complete in [32] and described in Appendix A. $\square$

Corollary 6.2. *Let H be a finite solvable group acting faithfully, irreducibly and primitively on a finite $\mathbb{F}_p$ -space V . If V contains no abundant vector, then H must be metacyclic.*

Proof. If H had a regular orbit on V , then some $v \in V$ would satisfy $H_v = 1$, and hence $\rho_H(v) = |H|$. Thus v would be abundant. Therefore H has no regular orbit. If H were not metacyclic, Theorem A.1(1) would again give an abundant vector. Hence H is metacyclic. $\square$

6.2. The metacyclic case. We now consider faithful irreducible primitive metacyclic groups. We first prove a theorem on regular pairs for primitive subgroups of $\Gamma\text{L}(1, Q)$, and then apply it to the metacyclic groups. Related semilinear estimates are used again in Lemma 8.5.

Let $Q = p^f$ and identify the natural f -dimensional $\mathbb{F}_p$ -space with the field $\mathbb{F}_Q$. Write $A = \mathbb{F}_Q^\times$ for the group of scalar transformations, and

$$\Gamma\text{L}(1, Q) = \{z \mapsto az^\sigma \mid a \in A, \sigma \in \text{Gal}(\mathbb{F}_Q/\mathbb{F}_p)\} = A \rtimes \text{Gal}(\mathbb{F}_Q/\mathbb{F}_p).$$

This is the full semilinear group of $\mathbb{F}_Q$. The notation $\Gamma(p^f)$ used in [12, 27, 29, 31] denotes the same group. For $K \leq \Gamma\text{L}(1, Q)$, let $n_{\text{reg}}(K)$ denote the number of regular orbits of the diagonal action of K on $\mathbb{F}_Q \times \mathbb{F}_Q$.

The natural homomorphism with kernel A is given by

$$\Gamma\text{L}(1, Q) \longrightarrow \text{Gal}(\mathbb{F}_Q/\mathbb{F}_p), \quad (z \mapsto az^\sigma) \longmapsto \sigma.$$

For a prime power q and a positive integer t , let $R(q, t)$ denote the number of regular orbits of $\text{Gal}(\mathbb{F}_{q^t}/\mathbb{F}_q)$ on $\mathbb{F}_{q^t}$.

Lemma 6.3. *Let $K \leq \Gamma\text{L}(1, Q)$, where $Q = p^f$, and set $U = K \cap A$. Let $C \cong K/U$ be the image of K in $\text{Gal}(\mathbb{F}_Q/\mathbb{F}_p)$. Write $t = |C|$ and $q = p^{f/t}$. Then*

$$n_{\text{reg}}(K) \geq |A : U| R(q, t).$$

Proof. The group $\text{Gal}(\mathbb{F}_Q/\mathbb{F}_p)$ is cyclic of order f . Consequently $t \mid f$, $Q = q^t$ and $C = \text{Gal}(\mathbb{F}_Q/\mathbb{F}_q)$. Thus C has $R(q, t)$ regular orbits on $\mathbb{F}_Q$. Let $\mathcal{R}$ be one of these orbits, and define

$$S_{\mathcal{R}} = \{(x, \xi x) : x \in A, \xi \in \mathcal{R}\}.$$

For $x \in A$ and $\xi \in \mathcal{R}$, suppose that an element $z \mapsto az^\sigma$ of K fixes both x and ξx . Then $ax^\sigma = x$ and $a(\xi x)^\sigma = \xi x$. Since σ is a field automorphism, $(\xi x)^\sigma = \xi^\sigma x^\sigma$. As a , x and x^σ are nonzero, the two equalities give

$$\frac{a\xi^\sigma x^\sigma}{ax^\sigma} = \frac{\xi x}{x},$$

and hence $\xi^\sigma = \xi$. Since $\mathcal{R}$ is regular, $\sigma = 1$, and then $ax = x$ forces $a = 1$. Hence $(x, \xi x)$ is a regular pair.

The set $S_{\mathcal{R}}$ is K -invariant: the map $z \mapsto az^\sigma$ sends $(x, \xi x)$ to $(ax^\sigma, \xi^\sigma ax^\sigma)$, where $ax^\sigma \in A$ and $\xi^\sigma \in \mathcal{R}$. Since $\mathcal{R}$ is regular and $|C| = t$, the orbit-stabiliser theorem gives $|\mathcal{R}| = t$. Hence $|S_{\mathcal{R}}| = |A|t$, and all its elements are regular. Each of its K -orbits therefore has $|K| = |U|t$ elements, so it contains $|A : U|$ orbits of regular pairs. The sets $S_{\mathcal{R}}$ arising from distinct choices of $\mathcal{R}$ are disjoint, and summing over the $R(q, t)$ choices proves the result. $\square$

Lemma 6.4. *Let q be a prime power and let $t \geq 1$ be an integer. Then $R(q, t) < 5$ only for*

$$(q, t) \in \{(2, 1), (2, 2), (2, 3), (2, 4), (3, 1), (3, 2), (4, 1)\}.$$

Proof. An element $\xi \in \mathbb{F}_{q^t}$ has trivial stabiliser in $\text{Gal}(\mathbb{F}_{q^t}/\mathbb{F}_q)$ precisely when it lies in no proper intermediate subfield, or equivalently when $\mathbb{F}_q(\xi) = \mathbb{F}_{q^t}$. For each divisor d of t , let N_d be the number of elements $\xi \in \mathbb{F}_{q^d}$ satisfying $\mathbb{F}_q(\xi) = \mathbb{F}_{q^d}$. Every element of $\mathbb{F}_{q^d}$ generates $\mathbb{F}_{q^e}$ over $\mathbb{F}_q$ for a unique divisor e of d , and hence

$$q^d = \sum_{e \mid d} N_e.$$

Möbius inversion gives

$$N_t = \sum_{d|t} \mu(t/d) q^d,$$

where μ is the Möbius function. Thus N_t is the number of elements with trivial stabiliser. Every regular orbit has length t , and therefore

$$R(q, t) = \frac{N_t}{t} = \frac{1}{t} \sum_{d|t} \mu(t/d) q^d.$$

For $t = 1$, we have $R(q, 1) = q$. As q is a prime power, $R(q, 1) < 5$ precisely when $q \in \{2, 3, 4\}$. For $t = 2, 3, 4$, respectively,

$$R(q, 2) = \frac{q^2 - q}{2}, \quad R(q, 3) = \frac{q^3 - q}{3}, \quad R(q, 4) = \frac{q^4 - q^2}{4}.$$

Thus $R(q, 2) < 5$ precisely when $q \in \{2, 3\}$, while $R(q, 3) < 5$ and $R(q, 4) < 5$ precisely when $q = 2$. For $t \geq 5$, every element of $\mathbb{F}_{q^t}$ that does not generate the extension over $\mathbb{F}_q$ lies in a subfield $\mathbb{F}_{q^e}$ with $e \mid t$ and $e \leq t/2$. Hence

$$tR(q, t) \geq q^t - \sum_{1 \leq e \leq \lfloor t/2 \rfloor} q^e.$$

For $q \geq 3$ the subtracted sum is below $q^t/2$, whence $tR(q, t) > q^t/2 \geq 5t$ and $R(q, t) \geq 5$. For $q = 2$ the same bound gives

$$tR(2, t) \geq 2^t - 2^{\lfloor t/2 \rfloor + 1} + 2.$$

The difference

$$2^t - 2^{\lfloor t/2 \rfloor + 1} + 2 - 5t$$

is positive for $t = 5, 6$ and increases thereafter separately along the odd and even values of t . Hence $R(2, t) \geq 5$ for all $t \geq 5$. $\square$

Theorem 6.5. *Let $K \leq \Gamma\text{L}(1, Q)$ act primitively on the $\mathbb{F}_p$ -space $\mathbb{F}_Q$. Then K has a regular pair, and has at least five orbits of regular pairs except for the six groups*

$$(Q, K) = (2, 1), (3, C_2), (4, \Gamma\text{L}(1, 4)), (8, \Gamma\text{L}(1, 8)), (9, \Gamma\text{L}(1, 9)), (16, \Gamma\text{L}(1, 16)),$$

whose values of $n_{\text{reg}}(K)$ are 4, 4, 1, 2, 3, 3 respectively.

Proof. Suppose $n_{\text{reg}}(K) < 5$. Set $U = K \cap A$, and let C be the image of K under the natural homomorphism to $\text{Gal}(\mathbb{F}_Q/\mathbb{F}_p)$. Let $\mathbb{F}_q$ be the subfield of $\mathbb{F}_Q$ fixed pointwise by C , and set $t = [\mathbb{F}_Q : \mathbb{F}_q]$. By Lemma 6.3, $|A : U| R(q, t) < 5$. Thus $R(q, t) < 5$, so (q, t) is one of the seven pairs in Lemma 6.4.

If $t = 1$, then $K = U \leq A$. For $q = 2$, the bound forces the trivial group on $\mathbb{F}_2$. For $q = 3$, it forces $U = A \cong C_2$. For $q = 4$, it forces $U = A \cong C_3$, but every ordered pair other than $(0, 0)$ is then regular. There are therefore $(16 - 1)/3 = 5$ orbits of regular pairs, contrary to $n_{\text{reg}}(K) < 5$. If $(q, t) = (2, 2)$, then $Q = 4$ and $A \cong C_3$. Suppose that $U = 1$. Then K has order 2, and its nonidentity element has the form $z \mapsto az^2$ for some $a \in \mathbb{F}_4^\times$. This element fixes a^{-1} , since $a(a^{-1})^2 = a^{-1}$. Thus $\{0, a^{-1}\}$ is a proper nonzero K -invariant $\mathbb{F}_2$ -subspace of $\mathbb{F}_4$, contrary to the primitivity of K . Therefore $U = A$ and $K = \Gamma\text{L}(1, 4)$. The pairs $(2, 3), (2, 4), (3, 2)$ correspond to $Q = 8, 16, 9$. In each, $|A : U| R(q, t) \leq n_{\text{reg}}(K) < 5$ forces $|A : U| < 5/R(q, t)$, and as $|A : U|$ divides $|A|$ this leaves only $|A : U| = 1$: for $Q = 8$, $|A| = 7$ and $R = 2$, whence $|A : U| < 5/2$. For $Q = 16$, $|A| = 15$ and $R = 3$, whence $|A : U| < 5/3$. For $Q = 9$, $|A| = 8$ and $R = 3$, whence $|A : U| < 5/3$. In every case, the next divisor of $|A|$ exceeds the bound. Hence $U = A$ and K is $\Gamma\text{L}(1, 8), \Gamma\text{L}(1, 16), \Gamma\text{L}(1, 9)$ respectively. This leaves exactly the six named groups. Their values of n_{reg} and the existence of a regular pair in each are verified in Appendix A.3. $\square$

Corollary 6.6. *Let W be a finite $\mathbb{F}_p$ -space and let $K \leq \text{GL}(W)$ be metacyclic, irreducible and primitive. Then K has a regular pair on W .*

Proof. A metacyclic group is abelian-by-nilpotent. By a theorem of Kovács [25, Theorem 5.3.1], K therefore normalises a Singer cycle. The normaliser of a Singer cycle is the semilinear group [25, Theorem 2.3.5]. Hence Theorem 6.5 applies. $\square$

Theorem 6.7. *Let H be a finite solvable group, and let V be a faithful irreducible primitive H -module. Then V contains an abundant vector.*

Proof. Suppose that V contains no abundant vector. By Corollary 6.2, H is metacyclic. Corollary 6.6 gives a regular pair, and Lemma 2.1 then gives an abundant vector, a contradiction. $\square$

7. LOCAL ALTERNATIVES FOR IMPRIMITIVE MODULES

Theorem 6.7 gives an abundant vector in every faithful irreducible primitive module. For an imprimitive module, however, an abundant vector for the group induced on one summand is not enough: Theorem 3.1 involves every bottom group arising from a coordinate chain. We therefore need a pair of local orbits that works uniformly along every coordinate chain, without assuming that the top group has a regular orbit on the power set of the index set. Such a regular orbit need not exist for an arbitrary solvable transitive group [11].

Definition 7.1. Let W be a finite vector space and let $J \leq \text{GL}(W)$ be solvable and primitive. We call an ordered pair $(\mathcal{O}_0, \mathcal{O}_1)$ of distinct J -orbits on W a *balanced pair* for J if, for every chain $L \trianglelefteq K \trianglelefteq J$, there exist $a \in \mathcal{O}_0$ and $b \in \mathcal{O}_1$ such that $\rho_L(a)\rho_L(b) \geq 1$. We call a J -orbit $\mathcal{O}$ on W a *chain-abundant orbit* if, for every chain $L \trianglelefteq K \trianglelefteq J$, there exists $a \in \mathcal{O}$ such that $\rho_L(a) \geq 1$. In both conditions, the orbits are fixed, but the representatives may vary with the chain.

Lemma 7.2. *Let W be a finite vector space and let $J \leq \text{GL}(W)$ be solvable and primitive. The following constructions give balanced pairs for J :*

- (1) *Any two distinct chain-abundant orbits $\mathcal{O}_0 \neq \mathcal{O}_1$ form a balanced pair.*
- (2) *If $v \in W$ is a nonzero regular vector, then $(v^J, \{0\})$ is a balanced pair.*
- (3) *If $(a, b) \in W \times W$ is a regular pair and a, b lie in distinct J -orbits, then (a^J, b^J) is a balanced pair.*

Proof. In (1), for every chain there exist $a \in \mathcal{O}_0$ and $b \in \mathcal{O}_1$ such that $\rho_L(a) \geq 1$ and $\rho_L(b) \geq 1$. Hence the product is at least 1. In (2), for the bottom group L of any chain the stabiliser $L_v = L \cap J_v = 1$. Hence $\rho_L(v) = |L|$ and $\rho_L(v)\rho_L(0) = |L|/\nu(L) \geq 1$. In (3), for the bottom group L of any chain we have $L_a \cap L_b = L \cap J_a \cap J_b = 1$. Hence (a, b) is a regular pair for L and $\rho_L(a)\rho_L(b) \geq 1$ by Lemma 2.1. In each case the two orbits are distinct. $\square$

Let W be a finite vector space and let $J \leq \text{GL}(W)$ be solvable and primitive. For each chain $L \trianglelefteq K \trianglelefteq J$, define $z_L = \rho_L(0) = 1/\nu(L)$ and $n_L = \max_{0 \neq w \in W} \rho_L(w)$. Thus z_L is the abundance ratio of the zero vector, while n_L is the largest abundance ratio among the nonzero vectors of W .

Theorem 7.3. *Let J be a nontrivial faithful irreducible primitive solvable linear group on W . Then either J has a balanced pair, or J is metacyclic with no regular vector, or the action of J on W is linearly equivalent to one of the nine residual actions of Theorem A.1(2). In the last case, $z_L n_L^3 \geq 1$ for every chain $L \trianglelefteq K \trianglelefteq J$.*

Proof. If J has a regular vector v , then Lemma 7.2(2) makes $(v^J, \{0\})$ a balanced pair. We may therefore suppose that J has no regular vector and is not metacyclic. By Theorem A.1(2), either J has a balanced pair, or its action on W is linearly equivalent to one of the nine residual actions and $z_L n_L^3 \geq 1$ for every chain $L \trianglelefteq K \trianglelefteq J$. $\square$

8. IMPRIMITIVE MODULES

Definition 8.1. Let V be a finite vector space and let $H \leq \text{GL}(V)$. A *system of imprimitivity* for H on V is a decomposition $V = W_1 \oplus \cdots \oplus W_m$ into nonzero subspaces such that H permutes $\{W_1, \dots, W_m\}$ transitively. The system is *nontrivial* if $m > 1$. It is *unrefinable* if there is no system of imprimitivity $V = U_1 \oplus \cdots \oplus U_n$ with $n > m$ such that every U_j is contained in some W_i .

Lemma 8.2. Let H be a finite group acting faithfully, irreducibly and imprimitively on a finite $\mathbb{F}_p$ -space V , and let $V = W_1 \oplus \cdots \oplus W_m$ be a nontrivial unrefinable system of imprimitivity. Let J be the group induced on W_1 by its setwise stabiliser H_{W_1} . Then J is nontrivial, faithful, irreducible and primitive.

Proof. The action of J on W_1 is faithful by definition. For each i , choose $t_i \in H$ with $W_1^{t_i} = W_i$, taking $t_1 = 1$.

Suppose first that $J = 1$. Then H_{W_1} fixes W_1 pointwise. Choose $0 \neq w_1 \in W_1$ and set $w_i = w_1^{t_i}$. If $t'_i \in H$ also carries W_1 to W_i , then $t_i(t'_i)^{-1} \in H_{W_1}$, and hence $w_1^{t_i} = w_1^{t'_i}$. Thus w_i is independent of the choice of t_i . If $W_i^h = W_j$, then both $t_i h$ and t_j carry W_1 to W_j , so $w_i^h = w_j$. Therefore H fixes $w = w_1 + \cdots + w_m$. Since $W_1 \oplus \cdots \oplus W_m$ is a direct sum, $w \neq 0$. As $m > 1$, the line $\mathbb{F}_p w$ is a proper H -submodule of V . This contradicts the irreducibility of V , so $J \neq 1$.

Suppose that $0 < U < W_1$ is J -invariant, and set $U_i = U^{t_i}$. The same argument shows that U_i is independent of the choice of t_i and that H permutes $U_1, \dots, U_m$. Hence $U_1 \oplus \cdots \oplus U_m$ is a proper nonzero H -submodule of V , again contradicting irreducibility. Therefore J is irreducible.

Finally, suppose that J is imprimitive, and let $W_1 = U_1 \oplus \cdots \oplus U_s$ be a nontrivial system of imprimitivity for J . For each i , transport this decomposition to W_i using t_i . The resulting decomposition of W_i is independent of the choice of t_i , since H_{W_1} permutes $U_1, \dots, U_s$. We obtain

$$V = \bigoplus_{i=1}^m \bigoplus_{j=1}^s U_j^{t_i}.$$

Since H is transitive on $\{W_1, \dots, W_m\}$ and J is transitive on $\{U_1, \dots, U_s\}$, the group H permutes the ms summands in this decomposition transitively. This is a system of imprimitivity that strictly refines $W_1 \oplus \cdots \oplus W_m$, contrary to unrefinability. Thus J is primitive. $\square$

For the next three lemmas, let H be a finite solvable group acting faithfully, irreducibly and imprimitively on a finite $\mathbb{F}_p$ -space V . Fix a nontrivial unrefinable system $V = W_1 \oplus \cdots \oplus W_m$, set $I = \{1, \dots, m\}$, where i indexes W_i , and let $P \leq \text{Sym}(I)$ be the group induced by H on I . Let J be the group induced on W_1 by its setwise stabiliser H_{W_1} . Then P is solvable and transitive. The group J is solvable, and Lemma 8.2 shows that it is nontrivial, faithful, irreducible and primitive.

For each $i \in I$, choose $t_i \in H$ with $W_1^{t_i} = W_i$, taking $t_1 = 1$. The map $w \mapsto w^{t_i^{-1}}$ from W_i to W_1 identifies the local group J_i with J . Under this identification, we write each coordinate chain as $L_i \trianglelefteq K_i \trianglelefteq J$, and abundance ratios are unchanged. In coordinate i , we use the same symbol $\mathcal{O}$ for the J -orbit on W_1 and its transported orbit $\mathcal{O}^{t_i}$ on W_i . For every chain $L \trianglelefteq K \trianglelefteq J$, set $z_L = \rho_L(0) = 1/\nu(L)$ and $n_L = \max_{0 \neq w \in W_1} \rho_L(w)$.

Lemma 8.3 (Balanced case). *If J has a balanced pair, then V contains an abundant vector.*

Proof. Let $(\mathcal{O}_0, \mathcal{O}_1)$ be a balanced pair for J . By Theorem 1.5 choose $\Gamma \subseteq I$ with $\rho_P(\Gamma) \geq 1$. Since P is transitive on I and $m > 1$, it is nontrivial and $\rho_P(\emptyset) = \rho_P(I) = 1/\nu(P) < 1$. Therefore Γ is neither empty nor all of I . For each coordinate $i \in I$, with bottom group L_i in its coordinate chain, pick representatives $a_i \in \mathcal{O}_0$ and $b_i \in \mathcal{O}_1$ with $\rho_{L_i}(a_i) \rho_{L_i}(b_i) \geq 1$. Consider two choices of local vectors. In the first, use a_i when $i \in \Gamma$ and b_i when $i \notin \Gamma$. In the second,

use b_i when $i \in \Gamma$ and a_i when $i \notin \Gamma$. The corresponding products of local ratios,

$$\prod_{i \in \Gamma} \rho_{L_i}(a_i) \prod_{i \notin \Gamma} \rho_{L_i}(b_i) \quad \text{and} \quad \prod_{i \in \Gamma} \rho_{L_i}(b_i) \prod_{i \notin \Gamma} \rho_{L_i}(a_i),$$

multiply to $\prod_{i \in I} \rho_{L_i}(a_i) \rho_{L_i}(b_i) \geq 1$. Hence at least one of them is ≥ 1 . Let x be the corresponding global vector. By Section 3, the entries chosen from the distinct J -orbits $\mathcal{O}_0$ and $\mathcal{O}_1$ have different colours. In the first choice, entries from $\mathcal{O}_0$ occur precisely at the coordinates indexed by Γ . In the second choice, they occur precisely at the coordinates indexed by Γ^c . In either case write $T = T_x$. It preserves Γ in the first choice and Γ^c in the second. Since $P_\Gamma = P_{\Gamma^c}$, we have $T \leq P_\Gamma$ in either case and, by (3.1), $|P : T|/\nu(T) \geq \rho_P(\Gamma) \geq 1$. The coordinate chain inequality (3.2) then gives $\rho_H(x) \geq (|P : T|/\nu(T)) \prod_i \rho_{L_i}(x_i) \geq 1$. $\square$

Lemma 8.4 (Cubic condition). *If $z_{L_i} n_{L_i}^3 \geq 1$ for every $i \in I$, then V contains an abundant vector.*

Proof. Apply Theorem 5.1 to P , taking $z_i = z_{L_i}$ and $n_i = n_{L_i}$ for each coordinate i , to obtain $\Gamma \subseteq I$ with $\rho_P(\Gamma) \prod_{i \in \Gamma} z_i \prod_{i \notin \Gamma} n_i \geq 1$. Place the zero vector on the summands indexed by Γ , and on every other summand choose a nonzero vector attaining n_{L_i} . Let x be their sum and write $T = T_x$. If $\Gamma = \emptyset$ or $\Gamma = I$, then $T \leq P_\Gamma = P$. Otherwise a zero entry and a nonzero entry have different colours, so again $T \leq P_\Gamma$. Hence Theorem 3.1 and (3.1) give

$$\rho_H(x) \geq \rho_P(\Gamma) \prod_{i \in \Gamma} z_i \prod_{i \notin \Gamma} n_i \geq 1. \quad \square$$

Lemma 8.5 (Semilinear case). *If J is metacyclic and has no regular vector in W_1 , then $z_L n_L^3 \geq 1$ for every chain $L \trianglelefteq K \trianglelefteq J$.*

Proof. Write $Q = |W_1|$. As in the proof of Corollary 6.6, we may identify W_1 with $\mathbb{F}_Q$ in such a way that

$$J \leq \Gamma L(1, Q) = \mathbb{F}_Q^\times \rtimes \text{Gal}(\mathbb{F}_Q/\mathbb{F}_p).$$

Here $Q \geq 4$, since for $Q \in \{2, 3\}$ every subgroup of $\Gamma L(1, Q)$ has a regular nonzero vector, whereas J has none. Fix a chain $L \trianglelefteq K \trianglelefteq J$, and let $A_L = L \cap \mathbb{F}_Q^\times$ be its scalar part and let C_L be the image of L under the natural homomorphism to $\text{Gal}(\mathbb{F}_Q/\mathbb{F}_p)$. Both A_L and C_L are cyclic, and L fits into the exact sequence

$$1 \longrightarrow A_L \longrightarrow L \longrightarrow C_L \longrightarrow 1,$$

which need not split.

For $v \neq 0$ no nonidentity scalar fixes v . Hence the stabiliser L_v meets $\mathbb{F}_Q^\times$ trivially and embeds in the cyclic group C_L . In particular, L_v is cyclic, $\nu(L_v) = |L_v|$ and $\rho_L(v) = |L|/|L_v|^2$. Writing $s_L = \min_{v \neq 0} |L_v|$ we obtain $n_L = |L|/s_L^2$, and since $\nu(L) \leq |L|$,

$$z_L n_L^3 = \frac{|L|^3}{s_L^6 \nu(L)} \geq \frac{|L|^2}{s_L^6}.$$

If $s_L = 1$ this yields $z_L n_L^3 \geq |L|^2 \geq 1$.

Assume now $s_L \geq 2$, so that every nonzero vector is fixed by a nonidentity element of L . Such an element has the form $z \mapsto az^\sigma$ with $\sigma \neq 1$, because a nonidentity scalar fixes no nonzero vector. Fix $\sigma \neq 1$ in C_L , and let $c = |\sigma|$. The scalar coefficients a for which the map $z \mapsto az^\sigma$ belongs to L form a single coset $a_\sigma A_L$ in $\mathbb{F}_Q^\times$, and this map fixes $v \neq 0$ precisely when $v^{1-\sigma} = a$, where $v^{1-\sigma} := v/v^\sigma$. For fixed σ , each v determines at most one element above σ that fixes it. Equivalently, the number of pairs (g, v) with g lying above σ and $0 \neq v \in W_1$ fixed by g is

$$|\{v \neq 0 : v^{1-\sigma} \in a_\sigma A_L\}| \leq |A_L|(Q^{1/c} - 1) \leq |A_L|(\sqrt{Q} - 1),$$

because the kernel of $v \mapsto v^{1-\sigma}$ is the multiplicative group of the fixed field $\mathbb{F}_Q^{(\sigma)}$, which has order $Q^{1/c} - 1$. For each nonzero v , exactly $|L_v| - 1$ nonidentity elements of L fix v , and $|L_v| - 1 \geq s_L - 1$. Counting the pairs (g, v) with $g \in L \setminus \{1\}$, $v \neq 0$ and $v^g = v$, first by v and

then by the image σ of g in C_L , gives

$$(Q-1)(s_L-1) \leq \sum_{v \neq 0} (|L_v|-1) = \sum_{\sigma \neq 1} |\{v \neq 0 : v^{1-\sigma} \in a_\sigma A_L\}| < |A_L||C_L|\sqrt{Q},$$

whence $|A_L| > (s_L-1)(Q-1)/(|C_L|\sqrt{Q})$. Using $|L| = |A_L||C_L|$ in the preceding bound gives $z_L n_L^3 \geq |L|^2/s_L^6$. Substituting the inequality for $|A_L|$ and using $(Q-1)^2 \geq Q^2/2$ for $Q \geq 4$, we obtain

$$z_L n_L^3 \geq \frac{|A_L|^2 |C_L|^2}{s_L^6} > \frac{(s_L-1)^2 (Q-1)^2}{Q s_L^6} \geq \frac{(s_L-1)^2}{2 s_L^6} Q,$$

which is at least 1 as soon as $Q \geq 2s_L^6/(s_L-1)^2$. Finally,

$$s_L \leq |C_L| \leq |\text{Gal}(\mathbb{F}_Q/\mathbb{F}_p)| = \log_p Q \leq \log_2 Q.$$

Hence the sufficient bound $Q \geq 2s_L^6/(s_L-1)^2$ can fail only when $2^{s_L} \leq Q < 2s_L^6/(s_L-1)^2$. The function $2s_L^6/(s_L-1)^2$ is increasing for $s_L \geq 2$. At $s_L = 18$ the lower bound 2^{s_L} is already at least this function, and the ratio $2^{s_L}(s_L-1)^2/(2s_L^6)$ increases for $s_L \geq 18$. Therefore $s_L \leq 17$ and $Q < 2 \cdot 17^6/16^2 < 2^{18}$. The finitely many summand sizes $Q < 2^{18}$ that remain are covered, for every possible bottom group L , by Proposition A.2. Hence $z_L n_L^3 \geq 1$ for every chain. $\square$

Theorem 8.6. *Let H be a finite solvable group, and let V be a faithful irreducible imprimitive H -module. Then V contains an abundant vector.*

Proof. Choose a nontrivial unrefinable system of imprimitivity $V = W_1 \oplus \cdots \oplus W_m$, and let J be the group induced on W_1 by H_{W_1} . The group J is solvable and, by Lemma 8.2, nontrivial, faithful, irreducible and primitive, so Theorem 7.3 applies. If J has a balanced pair, Lemma 8.3 gives an abundant vector in V . If J is metacyclic with no regular vector, Lemma 8.5 gives $z_L n_L^3 \geq 1$ for every coordinate chain. If the action of J on W_1 is linearly equivalent to one of the nine residual actions, the same inequality follows from Theorem 7.3. In either of the last two cases, Lemma 8.4 gives an abundant vector in V . $\square$

9. PROOF OF THE ABUNDANCE THEOREM AND OF GLUCK'S CONJECTURE

The abundance theorem now follows immediately.

Proof of Theorem 1.4. By Corollary 2.4, it is enough to consider a faithful irreducible $\mathbb{F}_p H$ -module. Such a module is either primitive or imprimitive, and Theorems 6.7 and 8.6 prove the result in the two cases. $\square$

Gluck's conjecture now follows at once. By Theorem 1.4 every faithful completely reducible module for a finite solvable group contains an abundant element. Hence Lemma 2.2 gives $|G : \mathbf{F}(G)| \leq b(G)^2$ for every finite solvable group G . This proves Theorem 1.2.

APPENDIX A. FINITE VERIFICATIONS FOR GLUCK'S CONJECTURE

This appendix describes the four finite calculations used in the proof. Holt and Yang [12] present a classification of the residual actions and supply matrix data for the actions in their list. The reconstruction in [32] finds twelve further $\text{GL}(8, 3)$ -conjugacy classes absent from those data and proves that the corrected classification is complete. Nine of the twelve additional classes are linearly primitive, and only these are relevant here. We verify only the abundance and local conditions required for the residual actions, together with the small permutation and semilinear calculations used earlier in the paper.

The package consists of the directories `code` and `outputs`, which must remain together. Run each program in Table A.1 from the `code` directory. No program requires the output of another. The program `verify_residual_actions.g` reads `residual_action_data.g`, which contains the matrix generators and the vectors used below for the 411 linearly primitive entries representing matrix groups. Of these, 402 come from the data accompanying [12]. The other nine are representatives of the nine additional linearly primitive classes found in [32]. Thus the

certificates are checked without repeating either classification calculation. A path beginning with `outputs/` below refers to the package's `outputs` directory.

TABLE A.1. GAP programs and the parts of the proof they verify.

program	calculation
<code>verify_permutation.g</code>	Lemmas 4.3 and 5.6(3)
<code>verify_semilinear.g</code>	the six exceptional field orders in Theorem 6.5
<code>verify_residual_actions.g</code>	the certificates for residual actions in Theorem A.1
<code>verify_semilinear_threshold.g</code>	the finite calculation in Proposition A.2

The four GAP calculations are practical to rerun. In our runs, their CPU times ranged from a few seconds to less than half an hour. Each used less than 700 MB of RAM.

The directory `outputs` contains a transcript of each complete GAP calculation in Table A.1. The transcript and program have the same base name. For example, `verify_semilinear.g` is recorded in `outputs/verify_semilinear.out`. Each transcript ends with a summary of its numerical conclusions. The matrix generators and vectors are also provided in `residual_action_data.m` in Magma form.

The corresponding Magma programs [1] have the same base names as the programs in Table A.1, with the extension `.magma`. They implement the same checks separately and are not used in the proof. The GAP and Magma programs for residual actions read the same stored matrix generators and vectors.

A.1. Representation of the groups and the computation of ν . All completed GAP calculations used in the proof were carried out in GAP 4.15.1 [7]. The permutation calculation uses the GAP libraries `primgrp` 4.0.1 and `transgrp` 3.6.5, obtaining groups through `PrimitiveGroup` and `TransitiveGroup`. The programs `verify_semilinear.g` and `verify_residual_actions.g` use `IRREDSOL` 1.4.4 for irreducibility and primitivity tests. Linear groups are represented as matrix groups over the prime field $\mathbb{F}_p$. The residual actions are constructed from the generators in `residual_action_data.g`. The semilinear groups $\Gamma L(1, Q)$ are constructed directly over $\mathbb{F}_p$.

To compute $\nu(X)$ exactly, the program sets $\nu(X) = |X|$ when X is nilpotent. Otherwise it calls `ConjugacyClassesSubgroups` on X and takes the greatest order among the representatives found to be nilpotent by `IsNilpotentGroup`. Every nilpotent subgroup of X is conjugate to one of these representatives, so this maximum is $\nu(X)$. The abundance ratio $\rho_X(x) = |X : X_x|/\nu(X_x)$ is then obtained from the orbit length and stabiliser, together with $\nu(X_x)$.

A.2. Small permutation groups. The program `verify_permutation.g` finds chain-abundant orbits on subsets of the required cardinalities for every representative of a solvable primitive permutation group of degree at most nine in GAP's library. For every such representative J on a domain D of degree at most nine, it forms every J -orbit on $\mathcal{P}(D)$, using the action on subsets, and every chain $L \trianglelefteq K \trianglelefteq J$, by letting K range over the normal subgroups of J and L over the normal subgroups of K . There are 21 such representatives in GAP's primitive groups library in degrees 2, ..., 9. This is exactly the family of chains over which Definition 4.1 quantifies. Because L need not be normal in J , the value $\rho_L(\Delta)$ can vary as Δ ranges over a single J -orbit, so examining a single orbit representative is not sufficient. For every group L occurring as the bottom group of a chain, the program examines members of the orbit until it finds one satisfying $|L : L_\Delta| \geq \nu(L_\Delta)$. If none does, every member is examined. The same bottom group arising from more than one chain is considered only once, since the inequality does not involve the intermediate group K . An orbit is chain-abundant precisely when a suitable member is found for every such L .

The proof needs chain-abundant orbits on subsets of prescribed sizes, not merely their total number. Of these 21 solvable primitive permutation groups, C_2 is handled separately. For each

of the remaining 20, the program verifies chain-abundant orbits on subsets of sizes 1 and 2, except for $\text{AFL}(1, 9)$ and $\text{AGL}(2, 3)$, where it verifies sizes 2 and 3. For those two groups the point stabiliser J_x , with $x \in D$, contains a nilpotent subgroup of order $16 > 9$, so the singleton orbit is not abundant for the chain $L = K = J$ and no chain-abundant orbit on singletons exists. For the group C_2 of degree two, the program verifies that there is exactly one chain-abundant orbit, the orbit of the singleton subsets. The conclusions in degrees 5, 7, 8, 9 are precisely those of Lemma 4.3. In degrees three and four the calculation independently confirms Lemma 4.2. The calculation in degree two confirms that the singleton orbit is the only chain-abundant orbit, the local fact used in Lemmas 4.6 and 5.7. For $(a_Z, a_N) = (2, 1)$, the condition $a_Z + 3a_N \leq |D|$ of Lemma 5.6(3) becomes $2 + 3 \cdot 1 \leq |D|$. For the two exceptional groups of degree nine, where $(a_Z, a_N) = (3, 2)$, it becomes $3 + 3 \cdot 2 \leq 9$. Thus the cubic pairs of that lemma are verified as well.

The program also verifies directly that all 501 representatives of solvable transitive permutation groups in GAP's `transgrp` library of degrees $2, \dots, 15$ have an abundant subset. This gives a finite check of Theorem 1.5 in those degrees, but is not used in its proof.

A.3. Small semilinear groups. The program `verify_semilinear.g` performs the finite calculation used in Theorem 6.5. For each of the six field orders $Q \in \{2, 3, 4, 8, 9, 16\}$ it constructs $\Gamma\text{L}(1, Q)$ as a matrix group over $\mathbb{F}_p$, using multiplication by a primitive element of $\mathbb{F}_Q$ and the Frobenius map $x \mapsto x^p$ as generators relative to a fixed basis over $\mathbb{F}_p$. It then enumerates the conjugacy classes of subgroups using `ConjugacyClassesSubgroups` and retains the solvable subgroups for which `IsIrreducibleMatrixGroup` and `IsPrimitiveMatrixGroup` both hold. These properties are invariant under conjugation, so class representatives suffice. For every retained subgroup K , the program computes $n_{\text{reg}}(K)$, the number of K -orbits of regular pairs defined in Subsection 6.2, by decomposing $\mathbb{F}_Q \times \mathbb{F}_Q$ into K -orbits and counting those with trivial stabiliser. It verifies that every proper retained subgroup $K < \Gamma\text{L}(1, Q)$ has at least five orbits of regular pairs. For the full semilinear groups, the values are

$$n_{\text{reg}}(\Gamma\text{L}(1, Q)) = 4, 4, 1, 2, 3, 3 \quad (Q = 2, 3, 4, 8, 9, 16).$$

In particular each value is at least 1, so each group possesses a regular pair, as Theorem 6.5 asserts.

A.4. Residual linear groups.

Theorem A.1 (Residual verification). *Let H be a finite solvable group acting faithfully, irreducibly and primitively on a finite $\mathbb{F}_p$ -space V . Suppose that H is not metacyclic and has no regular orbit. Then both of the following hold.*

- (1) *The module V contains an abundant vector.*
- (2) *Either H has a balanced pair, or its action on V is linearly equivalent to one of nine residual actions that are transitive on $V \setminus \{0\}$ and satisfy the cubic condition $z_L n_L^3 \geq 1$ for every chain $L \trianglelefteq K \trianglelefteq H$.*

Proof. By Theorem 6.1, every action satisfying the hypotheses is linearly equivalent to one of the actions represented by the 411 linearly primitive entries described at the beginning of this appendix. It is therefore enough to verify the action represented by each entry. These 411 entries represent 353 conjugacy classes in their respective general linear groups, so repetitions merely repeat some of the verifications. The program `verify_residual_actions.g` checks the matrix degree, ground field, solvability, irreducibility and primitivity before checking the certificate for each entry.

Abundant vectors. Fix one of the 411 entries, let $V = \mathbb{F}_p^d$, and let $J \leq \text{GL}(V)$ be the matrix group represented by that entry. We exhibit a vector v with $\rho_J(v) = |J : J_v|/\nu(J_v) \geq 1$ and verify this inequality directly. The calculation uses one of the following three criteria:

- (R) *regular:* if $|J_v| = 1$ then $\rho_J(v) = |J| \geq 1$.

(O) *order bound*: if $|J : J_v| \geq |J_v|$ then, since $\nu(J_v) \leq |J_v|$,

$$\rho_J(v) \geq \frac{|J : J_v|}{|J_v|} \geq 1.$$

This uses only the orbit length and the stabiliser order, and does not require $\nu(J_v)$ to be computed.

(E) *exact calculation of ν* : otherwise compute $\nu(J_v)$ by the method of Appendix A.1 and verify $|J : J_v| \geq \nu(J_v)$ directly.

The program constructs distinct points of the orbit v^J until the complete orbit has been obtained or until the following bound proves criterion (O). If $\lfloor \sqrt{|J|} \rfloor + 2$ distinct points have been found, the orbit has length greater than $\sqrt{|J|}$, so $|J : J_v| > \sqrt{|J|} > |J_v| \geq \nu(J_v)$ and criterion (O) already applies. The calculation therefore stops at that point. This bound makes the computation feasible: several residual modules are large — already $|V| = 3^{16} = 43\,046\,721$ occurs — so a full orbit enumeration is impractical, whereas this calculation constructs only slightly more than $\sqrt{|J|}$ orbit points. Only if the whole orbit has been obtained before this bound is reached does the program compute the exact stabiliser and, where necessary, $\nu(J_v)$. For the ten groups requiring criterion (E), the resulting stabilisers are small enough for their conjugacy classes of subgroups to be computed.

The file `residual_action_data.g` lists the vectors explicitly. They are not searched for during the verification. The file contains one certificate for each of the 411 linearly primitive entries. Matrix and vector entries in $\mathbb{F}_p$ are written as integers from 0 to $p-1$. The file contains no orbit lengths, stabilisers or values of ν . The program computes the information required by the applicable criterion. By the classification, none of the 411 listed actions has a regular orbit, so criterion (R) does not occur. For 401 entries the listed vector is verified by (O), and for the remaining 10 it is verified by (E). This proves item (1).

Balanced pairs and cubic conditions. Item (2) is used in Theorem 7.3. In the case of a regular pair, the vectors $a, b \in V$ satisfy $J_a \cap J_b = 1$ and $b \notin a^J$. Thus (a, b) is a regular pair with entries in distinct orbits, and (a^J, b^J) is balanced by Lemma 7.2(3). The program computes $J_a \cap J_b$ as the stabiliser of b in the previously computed stabiliser J_a , and verifies $b \notin a^J$ by testing membership in the explicitly enumerated orbit a^J .

The program uses the two vectors listed in `residual_action_data.g` for 394 entries. For eight further entries, all on modules of order at most 81, the same file lists vectors in two distinct nonzero orbits. The program enumerates those two orbits and, for each group L occurring as the bottom group of a chain $L \trianglelefteq K \trianglelefteq J$, verifies that each orbit contains a vector a with $\rho_L(a) \geq 1$. The two orbits are therefore chain-abundant and form a balanced pair by Lemma 7.2(1). Finally, the groups represented by the remaining nine entries are transitive on the nonzero vectors of their modules, so $\{0\}$ and $V \setminus \{0\}$ are their only orbits. For each of them the program enumerates every nonzero vector and verifies the *cubic condition*

$$z_L n_L^3 = \frac{(\max_{v \neq 0} \rho_L(v))^3}{\nu(L)} \geq 1 \quad \text{for every chain } L \trianglelefteq K \trianglelefteq J,$$

which is the hypothesis of Lemma 8.4. These nine actions are listed in Table A.2. Its first column gives the row, type and entry number in the corresponding Holt–Yang data file [12].

TABLE A.2. Residual actions for which the cubic condition is used.

Holt–Yang row and entry	V	$ J $
$19(E^-) : 2$	$\mathbb{F}_3^4$	640
$19(E^-) : 5$	$\mathbb{F}_3^4$	320
$19(E^-) : 6$	$\mathbb{F}_3^4$	160
$62 : 1$	$\mathbb{F}_3^2$	48
$62 : 2$	$\mathbb{F}_3^2$	24
$63 : 1$	$\mathbb{F}_5^2$	96
$63 : 2$	$\mathbb{F}_5^2$	48
$64 : 1$	$\mathbb{F}_7^2$	144
$66 : 1$	$\mathbb{F}_{11}^2$	240

Of the nine, entry $62 : 2$ can be checked by hand. Here $J \cong \mathrm{SL}(2, 3)$ acts on $V = \mathbb{F}_3^2$ and $\nu(J) = 8$, attained by the quaternion subgroup Q_8 . The group J is transitive on the eight nonzero vectors with point stabiliser of order 3, so every nonzero v has $\rho_J(v) = (24/3)/\nu(C_3) = 8/3$, whence $n_J = 8/3$ and $z_J n_J^3 = (1/8)(8/3)^3 = 64/27 \geq 1$. For the remaining possible bottom groups, the calculation is immediate: Q_8 acts regularly on the eight nonzero vectors, so $z_L n_L^3 = 8^3/8 = 64$ for $L = Q_8$, and its subgroups have trivial point stabilisers, giving 16 for $L = C_4$, 4 for $L = C_2$ and 1 for $L = 1$. Every chain $L \trianglelefteq K \trianglelefteq J$ therefore satisfies the cubic condition, with the minimum value 1 attained when $L = 1$. When checking chain-abundant orbits and the cubic condition, the program examines every subgroup L that can occur as the bottom group of a chain $L \trianglelefteq K \trianglelefteq J$. Since the inequalities do not involve the intermediate group K , this covers every coordinate chain that can arise in Theorem 3.1. No separate enumeration of chains is needed for regular pairs, because Lemma 7.2(3) applies to every chain. The program verifies regular pairs for 394 entries, chain-abundant orbits for eight entries, and the cubic condition for nine entries. These numbers sum to 411, proving (2). $\square$

A.5. The semilinear threshold.

Proposition A.2 (Semilinear subgroups below the threshold). *Let $Q < 2^{18}$ be a prime power and let $L \leq \Gamma\mathrm{L}(1, Q)$ act on $\mathbb{F}_Q$. Set $z_L = 1/\nu(L)$ and $n_L = \max_{v \neq 0} \rho_L(v)$. Then $z_L n_L^3 \geq 1$.*

Proof. Write $Q = p^f$. If $f = 1$, then $L \leq \Gamma\mathrm{L}(1, p) = \mathbb{F}_p^\times$ is cyclic and every nonzero vector has trivial stabiliser. Hence $z_L n_L^3 = |L|^2 \geq 1$. Assume that $f \geq 2$. The program `verify_semilinear_threshold.g` treats all 149 prime powers $Q = p^f < 2^{18}$ of this form.

Choose a generator ζ of $\mathbb{F}_Q^\times$ and identify ζ^t with t modulo $Q - 1$. With actions on the right,

$$\Gamma\mathrm{L}(1, Q) = \langle s, \varphi \mid s^{Q-1} = \varphi^f = 1, s^\varphi = s^p \rangle,$$

where s sends t to $t + 1$ and φ sends t to pt . Every subgroup has a unique description

$$L(u, r, j) = \langle s^u, \varphi^r s^j \rangle,$$

where $u \mid Q - 1$, $r \mid f$, $0 \leq j < u$, and $u \mid jT_{r,c}$, with $c = f/r$ and $T_{r,c} = 1 + p^r + \cdots + p^{(c-1)r}$. Indeed, $\langle s^u \rangle = L \cap \langle s \rangle$, the image of L in the Galois quotient is generated by the image of φ^r , and $(\varphi^r s^j)^c = s^{jT_{r,c}}$. Thus the divisibility condition is necessary and sufficient, and

$$|L(u, r, j)| = \frac{Q-1}{u} c.$$

The program runs through all parameters satisfying these conditions and therefore includes every subgroup.

The $\langle s^u \rangle$ -orbits on $\mathbb{F}_Q^\times$ correspond to the residue classes modulo u , each of length $(Q-1)/u$. The element $\varphi^r s^j$ acts on these classes as

$$\alpha(t) = p^r t + j \pmod{u}.$$

For each $h \mid c$, the fixed points of α^h are the solutions of

$$(p^{rh} - 1)t \equiv -j(1 + p^r + \cdots + p^{(h-1)r}) \pmod{u}.$$

This congruence has either no solution or $\gcd(p^{rh} - 1, u)$ solutions. Subtracting the points of smaller least period determines the greatest cycle length ℓ_L of α . Hence the greatest L -orbit length is

$$m_L = \frac{Q-1}{u} \ell_L.$$

If $|v^L| = m$, then L_v embeds in the cyclic Galois quotient. Thus $\nu(L_v) = |L_v| = |L|/m$, and consequently

$$\rho_L(v) = \frac{m^2}{|L|} \quad \text{and} \quad n_L = \frac{m_L^2}{|L|}.$$

Since $\nu(L) \leq |L|$, the required inequality follows whenever $n_L^3 \geq |L|$. The program finds only three nontrivial subgroups for which $n_L^3 < |L|$. For these groups it computes $\nu(L)$ from an isomorphic polycyclic presentation:

Q	(u, r, j)	$ L $	m_L	$\nu(L)$	$n_L^3/\nu(L)$
4	(1, 1, 0)	6	3	3	9/8
8	(1, 1, 0)	21	7	7	49/27
16	(1, 1, 0)	60	15	15	225/64

All three ratios exceed 1. The program finds one nontrivial equality case, with $Q = 9$, $(u, r, j) = (2, 1, 0)$, $|L| = \nu(L) = 8$, $m_L = 4$ and $n_L = 2$. Altogether the program enumerates 1896985 nontrivial subgroups. The trivial subgroup also gives equality. $\square$

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REFERENCES

- [1] W. Bosma, J. Cannon, and C. Playoust, *The Magma algebra system. I. The user language*, J. Symbolic Comput. **24** (1997), 235–265.
- [2] J. P. Cossey, Z. Halasi, A. Maróti, and H. N. Nguyen, *On a conjecture of Gluck*, Math. Z. **279** (2015), 1067–1080.
- [3] D. A. Craven, *On the irreducible character degrees of symmetric groups and their multiplicities*, Exp. Math. (2025), published online.
- [4] S. Dolfi, *Large orbits in coprime actions of solvable groups*, Trans. Amer. Math. Soc. **360** (2008), no. 1, 135–152.
- [5] S. Dolfi and E. Jabara, *Large character degrees of solvable groups with abelian Sylow 2-subgroups*, J. Algebra **313** (2007), 687–694.
- [6] A. Espuelas, *Large character degrees of groups of odd order*, Illinois J. Math. **35** (1991), no. 3, 499–505.
- [7] The GAP Group, *GAP – Groups, Algorithms, and Programming, Version 4.15.1*, 2025, available at www.gap-system.org.
- [8] W. Gaschütz, *Über die Φ -Untergruppe endlicher Gruppen*, Math. Z. **58** (1953), 160–170.
- [9] D. Gluck, *Trivial set-stabilizers in finite permutation groups*, Canad. J. Math. **35** (1983), 59–67.
- [10] D. Gluck, *The largest irreducible character degree of a finite group*, Canad. J. Math. **37** (1985), 442–451.
- [11] D. Gluck, *Set-stabilizers in solvable permutation groups*, J. Group Theory **28** (2025), 1297–1305.

- [12] D. F. Holt and Y. Yang, *Regular orbits of finite primitive solvable groups, the final classification*, J. Symbolic Comput. **116** (2023), 139–145.
- [13] N. N. Hung and Y. Yang, *Abelian subgroups, nilpotent subgroups, and the largest character degree of a finite group*, J. Algebra **551** (2020), 285–300.
- [14] I. M. Isaacs, *Character Theory of Finite Groups*, Academic Press, New York, 1976.
- [15] T. M. Keller and Y. Yang, *Regular and p -regular orbits of solvable linear groups*, J. Algebra **398** (2014), 509–518.
- [16] O. Manz and T. R. Wolf, *Representations of Solvable Groups*, London Math. Soc. Lecture Note Ser., vol. 185, Cambridge Univ. Press, Cambridge, 1993.
- [17] H. Meng, A. Ballester-Bolínches, and R. Esteban-Romero, *On large orbits of subgroups of linear groups*, Trans. Amer. Math. Soc. **372** (2019), no. 4, 2589–2612.
- [18] H. Meng, A. Ballester-Bolínches, and R. Esteban-Romero, *On large orbits of supersoluble subgroups of linear groups*, J. London Math. Soc. (2) **101** (2020), 490–504.
- [19] H. Meng and X. Guo, *On Gluck's conjecture for wreath product type groups*, J. Group Theory **28** (2025), 179–191.
- [20] A. Moretó, *The average character degree of finite groups and Gluck's conjecture*, J. Group Theory **26** (2023), 803–815.
- [21] A. Moretó and T. R. Wolf, *Orbit sizes, character degrees and Sylow subgroups*, Adv. Math. **184** (2004), 18–36.
- [22] L. Sabatini, *On stabilizers in finite permutation groups*, Bull. London Math. Soc. **58** (2026), e70201.
- [23] Á. Seress, *The minimal base size of primitive solvable permutation groups*, J. London Math. Soc. (2) **53** (1996), 243–255.
- [24] Á. Seress, *Primitive groups with no regular orbits on the set of subsets*, Bull. London Math. Soc. **29** (1997), 697–704.
- [25] M. W. Short, *The Primitive Soluble Permutation Groups of Degree Less Than 256*, Lecture Notes in Mathematics **1519**, Springer, Berlin, 1992.
- [26] T. R. Wolf, *Large orbits of supersoluble linear groups*, J. Algebra **215** (1999), 235–247.
- [27] Y. Yang, *Regular orbits of finite primitive solvable groups*, J. Algebra **323** (2010), 2735–2755.
- [28] Y. Yang, *Large character degrees of solvable $3'$ -groups*, Proc. Amer. Math. Soc. **139** (2011), no. 9, 3171–3173.
- [29] Y. Yang, *Regular orbits of finite primitive solvable groups, II*, J. Algebra **341** (2011), 23–34.
- [30] Y. Yang, *On Gluck's conjecture*, Bull. Sci. Math. **183** (2023), 103235.
- [31] Y. Yang, A. Vasil'ev, and E. Vdovin, *Regular orbits of finite primitive solvable groups, III*, J. Algebra **590** (2022), 139–154.
- [32] B. Zhang, *Regular orbits of finite quasi-primitive solvable linear groups: reconstruction and corrections*, preprint.
- [33] J. Zhang, *Finite groups with few regular orbits on the power set*, Algebra Colloq. **4** (1997), no. 4, 471–480.

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